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Budget-Aware Large Language Model Calls for MPaGE-Based Multi-Objective Heuristic Design

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Abstract

The `mpage_bmab` project studies how a finite budget of large language model (LLM) calls can be allocated more effectively in automatic heuristic design for multi-objective combinatorial optimization. The project builds on MPaGE, a framework that combines LLM-based program generation, SEMO-style multi-objective evolution, Pareto Front Grid selection, and semantic clustering of generated heuristics. The extension developed in this project replaces the largely fixed scheduling policy with a two-level budgeted multi-armed bandit controller: the first level selects the variation operator, and the second level selects the semantic cluster to exploit in the current generation. The implementation also introduces explicit budget accounting, Page–Hinkley drift detection, bounded reward computation, reward alignment with final managed-population hypervolume, and the area under the budget curve (AUBC) as a budget-efficiency metric.

This thesis synthesizes the theoretical background, system architecture, implementation details, and experimental evidence available in the project repository. The current result set contains 320 runs over four setups, four benchmark problem families, four LLM-call budgets, and five random seeds. The analyzed setups include the official Final-HV reward method, two reward-mode variants named Dense reward and Hybrid reward, and the budget-wrapped MPaGE-orig baseline. The empirical evidence indicates that the BMAB-style methods substantially improve AUBC over MPaGE-orig, while the Final-HV reward method provides the strongest final-HV ranking and Hybrid reward provides the strongest valid-yield behavior. Because each task–budget cell contains five seeds, the thesis interprets the findings as practical empirical evidence rather than as strong statistical proof.

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Chapter 1

Introduction

1.1 Research Context

Multi-objective combinatorial optimization is a central problem class in computer science. It appears in routing, scheduling, resource allocation, logistics, packing, and system design. Unlike single-objective optimization, the goal is not to identify one globally best solution under a scalar criterion. Instead, the algorithm must approximate a set of Pareto-optimal trade-offs, where improving one objective may necessarily degrade another. This requirement makes the design of effective search heuristics substantially more difficult, particularly when the feasible space is discrete and grows combinatorially with the problem size.

In many multi-objective combinatorial optimization problems (MOCOPs), the effectiveness of a heuristic depends strongly on the structure of the target task. A neighborhood rule that works well for a bi-objective traveling salesman problem may not transfer directly to a capacitated vehicle routing problem or a bi-objective knapsack problem. This task dependence motivates automatic heuristic design: instead of manually specifying every search rule, a system can generate candidate heuristic programs, execute them in a controlled evaluator, and retain the programs that produce stronger Pareto-front approximations.

Recent work has explored the use of large language models (LLMs) as program generators for this purpose. The MPaGE framework proposes Pareto-Grid-Guided LLMs for heuristic design in multi-objective combinatorial optimization [1]. Its main mechanism combines LLM-generated heuristic code, SEMO-style multi-objective evolution, Pareto Front Grid (PFG) selection, and LLM-based semantic clustering. The present project, `mpage_bmab`, is built on this foundation but addresses a narrower resource-allocation question: when the number of LLM calls is limited, how should the system allocate those calls to obtain useful heuristic populations earlier and more reliably?

This question is important because LLM-based heuristic design differs from ordinary evolutionary search in both cost structure and failure behavior. In a classical evolutionary

algorithm, generating one additional mutation can often be treated as a relatively cheap operation compared with evaluating the population. In an LLM-driven system, however, each candidate heuristic requires a model call, prompt construction, response parsing, code extraction, execution in a sandboxed evaluator, and population update. The generated program may be syntactically valid but semantically useless, or it may fail because it violates a task-specific contract. Therefore, the design of the outer search policy is not only an algorithmic preference; it directly determines how a scarce generative resource is converted into validated executable heuristics.

The project also has a broader research relevance within generative AI. Many applications of LLMs use a generate–evaluate–select loop, but the scheduling of generation calls is often treated as a fixed engineering detail. In contrast, this thesis studies generation as a sequential decision process. The LLM is not fine-tuned, and it is not used as an autonomous multi-agent planner. Instead, it acts as a stochastic program generator embedded inside an optimization loop whose state, rewards, and budget consumption are explicitly measured. This framing makes the project a concrete example of how generative models can be combined with classical decision-theoretic control.

1.2 Problem Statement

The repository documentation identifies budget allocation as a practical limitation of the baseline MPaGE workflow. MPaGE already provides a productive mechanism for generating and evolving heuristic programs, but its scheduling policy is comparatively static. In each generation, the system selects heuristic elites, clusters them semantically, applies mutation or crossover prompts, evaluates the resulting code, and updates the heuristic population. This process is effective as a heuristic-design loop, but it does not explicitly learn which operator or semantic region is currently yielding the most valuable improvement per LLM call.

This limitation matters because an LLM call is not a free computational step. It consumes API budget, wall-clock time, and experimental opportunity. Moreover, not every generated program is valid: some candidates fail to parse, violate the expected function signature, time out, or return infeasible outputs during evaluation. Therefore, the outer loop of LLM-driven heuristic design must be treated not only as an optimization process over generated programs, but also as a sequential decision problem under a hard budget.

The central problem studied in this thesis is therefore the following:

> How can an MPaGE-style LLM heuristic-design system allocate a fixed budget of generation and clustering calls so that the managed heuristic population achieves high hypervolume earlier, while preserving the final quality and diversity of the heuristic Pareto front?

The problem is deliberately formulated at the level of *heuristic-population search*. Each generated program is evaluated by an inner solver on fixed benchmark instances, but the proposed method does not directly optimize tours, vehicle routes, or knapsack vectors. Instead, it optimizes a population of heuristic programs that can produce such solutions. This creates a two-level structure: the inner level measures how well a heuristic solves a benchmark problem, while the outer level decides which heuristic programs should survive and which future generation calls should be made. Confusing these two levels can lead to incorrect interpretations of the results, so the thesis consistently distinguishes inner solution-level quality from outer heuristic-population quality.

1.3 Research Questions

The thesis is organized around four research questions.

1. **Budget efficiency.** Does a budget-aware scheduling policy improve the area under the outer hypervolume-vs-budget curve compared with the budget-wrapped MPaGE baseline?

2. **Terminal quality.** Does improving budget efficiency also preserve or improve final heuristic-population hypervolume, or does it merely move progress earlier without changing the final population?

3. **Reward design.** How do immediate dense HVI reward, final managed-population HV reward, and their hybrid differ in practice when all other finalized implementation components are held fixed?

4. **Validity and diagnostics.** To what extent do validity-related metrics, such as valid heuristic yield and invalid/null-score rate, explain differences in AUBC and final hypervolume?

These questions are narrower than a full re-evaluation of MPaGE across every table in the original paper. They focus on the budgeted outer search process implemented in `mpage_bmab`. This choice is intentional. The most distinctive contribution of the project is not a new benchmark task or a new LLM prompt family, but a controller that decides how to spend a fixed call budget across operators and semantic regions.

1.4 Research Motivation

The motivation for the project is supported by three observations found directly in the repository.

First, the implementation treats LLM calls as a limited resource. The module `mpage_bmab/budget.py` defines a `BudgetTracker` that records budget consumption through explicit `charge` operations. Both the proposed method in `mpage_bmab/main.py` and the baseline wrapper in `mpage_bmab/mpage_orig.py` receive a `--budget` argument so that methods can be compared under a common call budget.

Second, generated heuristics can be invalid or uninformative. The evaluator in `llm4ad/base/evaluate.py` executes generated programs in a separate process and may return a null score when the code is syntactically invalid, semantically incompatible with the task, or exceeds the allowed runtime. The validity analysis reported in Chapter 5 confirms that validity is an empirical phenomenon worth measuring, not merely an implementation detail.

Third, terminal hypervolume alone is insufficient for evaluating a budget-constrained LLM search process. The document `mpage_bmab/documents/METRICS_AUBC_HV.md` defines AUBC, the area under the hypervolume-vs-budget curve, to measure how quickly a method converts consumed budget into heuristic-population quality. Two methods may reach similar final hypervolume, but the method that reaches high-quality fronts earlier is more useful when LLM calls are expensive or when a run may be stopped before exhausting a large budget.

These observations motivate a controller that learns from the outcomes of previous generation calls. If crossover in one generation produces valid and diverse children, it should become more attractive; if a semantic cluster becomes saturated and no longer improves the population, it should receive fewer calls. Conversely, unexplored operators or clusters should not be abandoned too early, because the apparent lack of reward may be caused by limited evidence. This is precisely the exploration–exploitation tension studied in multi-armed bandits. The novelty in this thesis is the adaptation of that idea to the MPaGE heuristic-design loop, where arms correspond to variation operators and semantic cluster–operator combinations rather than ordinary solution-space moves.

The motivation is also practical for experimental research. LLM calls can make large grids of experiments expensive and time-consuming. A method that reaches a competitive heuristic population using fewer useful calls can make iterative research faster, especially during ablation studies, prompt debugging, and preliminary task exploration. AUBC is therefore not merely a secondary metric; it is a way of measuring whether the system makes meaningful progress throughout the budget horizon.

1.5 Research Objectives

The thesis has five concrete objectives.

1. Analyze the MPaGE baseline and identify the components retained in `mpage_bm`

ab, including the task templates, prompt families, semantic clustering, PFG selection, and secure evaluation infrastructure. The main supporting sources are `README.md`, `llm4ad/method/LLMPFG/`, and `mpage_bmab/documents/RETAINED_FROM_MPAGE.md`.

2. Present the proposed BMAB-LLM method as a budget-aware extension of MPaGE. The discussion covers hard budget accounting, operator-level bandit scheduling, cluster-level bandit scheduling, Page–Hinkley drift detection, final-HV-oriented reward computation, budget-aware exploration annealing, and final population flushing. The main implementation sources are `mpage_bmab/bmab_llm.py`, `mpage_bmab/bandit.py`, `mpage_bmab/reward.py`, and `mpage_bmab/cluster_manager.py`.

3. Explain the system implementation, benchmark evaluators, experimental harness, and reproducibility artifacts. The main sources are `mpage_bmab/main.py`, `mpage_bmab/mpage_orig.py`, `mpage_bmab/experiments/run.py`, and `mpage_bmab/experiments/aggregate.py`.

4. Analyze the available experimental evidence summarized in the thesis tables and figures, including AUBC, final hypervolume, invalid/null-score rate, and valid heuristic yield per budget.

5. Draw evidence-controlled conclusions. Claims are made only when they are supported by repository code, documentation, or experimental artifacts. When information is absent, the thesis explicitly states that no information was found in the project source code or documentation.

1.6 Research Scope

The empirical scope of this thesis is the analyzed comparison matrix represented by the thesis tables and figures. The current complete matrix contains four thesis-facing setups: Final-HV reward, Dense reward, Hybrid reward, and MPaGE-orig. The official proposed method is Final-HV reward, which corresponds to the implementation setup identifier `full`; the two reward-mode variants keep the finalized implementation but change the reward quality signal; and MPaGE-orig is the original MPaGE workflow wrapped under the same budget-recording protocol. The evaluated tasks are Bi-TSP, Tri-TSP, Bi-CVRP, and Bi-KP. The LLM-call budgets are 25, 50, 100, and 200. Each task–budget–setup cell is run with five seeds, from 2025 to 2029, as configured in `mpage_bmab/experiments/configs.py`.

Several additional ablation variants are implemented in the codebase, including `no_budget_anneal`, `no_ph`, `no_diversity`, `op_only`, and `cluster_only`. These

variants are important for future causal analysis, but they are not all present as complete result matrices in the current experimental directory. Consequently, Chapter 5 analyzes the four complete thesis-facing setups and treats the remaining unexecuted or incomplete ablations as future work rather than as established empirical findings.

The thesis does not re-evaluate the benchmark tables reported in the original MPaGE paper. The available result files measure outer heuristic-population hypervolume and AUBC under the `mpage_bmab` experimental harness. They should therefore be interpreted as evidence about budget-aware heuristic-population search, not as a direct reproduction of all normalized inner-solver metrics in the original paper.

The repository does not provide evidence of LLM fine-tuning, LoRA, PEFT, retrieval-augmented generation, diffusion models, GANs, knowledge distillation, or multi-agent autonomous planning. These topics are therefore not claimed as project contributions.

1.7 Research Methodology

The thesis follows an evidence-based repository analysis methodology. First, README files, design documents, implementation modules, evaluator definitions, scripts, result files, and generated figures were inspected to reconstruct the research problem and implementation architecture. Second, the mathematical and algorithmic foundations were organized around multi-objective optimization, hypervolume, LLM-based program synthesis, adaptive operator selection, budgeted multi-armed bandits, and online drift detection. Third, the result CSV and JSON files were analyzed using the metrics already defined by the project, particularly AUBC and final hypervolume.

The primary empirical metric is AUBC because it directly evaluates budget efficiency. Final hypervolume is used as a complementary terminal-quality metric. Valid heuristic yield and invalid/null-score rate are used as diagnostic indicators to understand whether improvements arise merely from producing more executable code or from allocating calls to more valuable regions of the heuristic-design search space. Chapter 5 reports normalized scores, mean ranks, cell-level pairwise wins, budget curves, and validity diagnostics generated from the result directory.

1.8 Thesis Organization

Chapter 2 reviews the theoretical background and related work: multi-objective combinatorial optimization, Pareto dominance, hypervolume, MPaGE, LLM-based program generation, multi-armed bandits, budgeted bandits, Page–Hinkley drift detection, and

AUBC. Chapter 3 presents the proposed BMAB-LLM method in detail, including the problem formulation, architecture, reward computation, bandit scheduler, and complete workflow. Chapter 4 describes the system implementation, benchmark evaluators, experimental harness, output artifacts, and reproducibility constraints. Chapter 5 analyzes the available experimental results and discusses the observed trends, limitations, and practical significance. Chapter 6 concludes the thesis and outlines future research directions.

Chapter 2

Background and Related Work

2.1 Multi-Objective Combinatorial Optimization

A multi-objective combinatorial optimization problem (MOCOP) can be written as

$$\min_{x \in \mathcal{X}} F(x) = \min_{x \in \mathcal{X}} \left(f_1(x), f_2(x), \ldots, f_m(x) \right), \tag{2.1}$$

where $\mathcal{X}$ is a discrete feasible set and m is the number of objectives. For two feasible solutions x and y, x Pareto-dominates y if $f_i(x) \leq f_i(y)$ for all objectives and $f_j(x) < f_j(y)$ for at least one objective. The set of non-dominated solutions forms a Pareto approximation, which represents different trade-offs rather than a single optimum.

The four benchmark families in the project instantiate this general setting. Bi-TSP and Tri-TSP evaluate tours under two or three distance objectives. Bi-CVRP evaluates route sets under distance-related objectives while enforcing vehicle-capacity constraints. Bi-KP evaluates binary item-selection vectors under two value objectives and a capacity constraint. In all cases, the inner solver maintains an archive of non-dominated candidate solutions and updates it when a generated heuristic proposes a feasible and non-dominated neighbor.

An important distinction in this project is that the LLM-generated program is not itself a final solution to the combinatorial problem. The generated code implements a function such as `select_neighbor`, which acts as a neighborhood operator inside an evaluator. The evaluator repeatedly calls this operator, updates an archive, and then computes the hypervolume of the resulting archive. This pattern appears in the project-local evaluator files under `mpage_bmab/_llm4ad/task/optimization/`.

2.2 Benchmark Problem Families

The benchmark problems used in the repository are deliberately diverse. They are all multi-objective and combinatorial, but they stress different properties of a generated heuristic. This diversity is important because a method that only performs well on one family may simply exploit a narrow representation bias rather than discovering generally useful heuristic-design behavior.

Bi-TSP and Tri-TSP are routing problems over complete graphs. A candidate solution is a permutation of cities, and a neighbor operator typically modifies the tour by swapping, reversing, inserting, or otherwise rearranging city positions. In Bi-TSP, the objective vector has two tour-cost components. In Tri-TSP, the same representation is evaluated under three distance matrices. The tri-objective version is harder from the perspective of the inner archive because a larger objective dimension generally increases the number of mutually non-dominated solutions. As a result, preserving useful diversity in the generated neighbor operator becomes more important.

Bi-CVRP introduces capacity-constrained route construction. A solution is not only an ordering of locations but a set of vehicle routes that must respect a capacity limit. A generated heuristic must therefore modify a structured route set while avoiding infeasible route loads. Compared with TSP, this task gives more opportunities for invalid or low-quality local moves, because changing the position of one customer can affect route feasibility and route balance.

Bi-KP uses a binary decision vector representing selected items. The objectives are value-related and the capacity constraint restricts the feasible subset of items. This problem has a different neighborhood geometry from routing: local moves often correspond to adding, removing, or exchanging selected items. A heuristic that performs well on routing tasks may not transfer directly to Bi-KP because permutation-based intuitions are less useful. Including Bi-KP therefore tests whether the outer heuristic-design process can adapt across representations.

All four tasks evaluate a generated heuristic on multiple fixed problem instances and aggregate its behavior into a two-dimensional outer score. The first score component is the negated mean inner hypervolume, and the second is mean runtime. This convention turns heuristic evaluation into a minimization problem: lower negated hypervolume is better, and lower runtime is better. The convention is simple but important, because it allows the same outer population manager to compare heuristics even when the inner problem has different objective dimensionalities.

2.3 Heuristic Design as a Two-Level Optimization Problem

The project can be understood as a two-level optimization process. At the inner level, a generated heuristic h is applied to a benchmark instance ξ to produce a set of candidate solutions:

$$A(h, \xi) = \text{Archive}\,(h, \xi)\,, \tag{2.2}$$

where $A(h, \xi)$ denotes the non-dominated archive returned by the evaluator after repeatedly applying the heuristic. The evaluator then computes an inner quality value, usually by hypervolume:

$$q(h, \xi) = HV_{\text{inner}}\,(A(h, \xi))\,. \tag{2.3}$$

Across a set of benchmark instances Ξ, the heuristic obtains an aggregated score:

$$s(h) = \left(-\frac{1}{|\Xi|} \sum_{\xi \in \Xi} q(h, \xi), \frac{1}{|\Xi|} \sum_{\xi \in \Xi} \tau(h, \xi) \right), \tag{2.4}$$

where $\tau(h, \xi)$ is the runtime of the inner evaluation. At the outer level, the algorithm maintains a population $\mathcal{H}$ of generated heuristics and seeks a high-quality Pareto set in the score space:

$$\mathcal{H}^\star \approx \text{ND}\{s(h) : h \in \mathcal{H}\}, \tag{2.5}$$

where ND denotes the non-dominated subset. This formulation clarifies why the thesis reports outer hypervolume and AUBC. The experiments do not directly compare individual tours or knapsack selections. They compare search processes that produce populations of heuristic programs.

This two-level view also explains the role of runtime. A heuristic that produces a strong inner archive but is consistently slow may be dominated by a slightly weaker but much faster heuristic. This is appropriate because automatic heuristic design should discover programs that are both effective and usable. The outer population therefore represents a trade-off between solution quality and computational cost.

2.4 Evolutionary Multi-Objective Optimization

Classical evolutionary multi-objective optimization provides the algorithmic context for MPaGE and BMAB-LLM. NSGA-II uses non-dominated sorting and crowding distance to maintain both convergence and diversity in a population [2]. MOEA/D decomposes a

multi-objective problem into scalar subproblems and optimizes them cooperatively [3]. SEMO-style algorithms use mutation and monotone archive updates to analyze and solve multi-objective pseudo-Boolean and combinatorial problems [4].

MPaGE inherits the archive-based and Pareto-based view of these methods, but changes the search object. Instead of evolving solution vectors directly, it evolves heuristic programs. A heuristic program is evaluated by running it inside a problem-specific solver and measuring the quality and runtime of the resulting Pareto archive. Thus, the outer search problem is itself multi-objective: it seeks heuristics that produce high inner-solver hypervolume while remaining computationally efficient.

This change of search object has methodological consequences. In a conventional evolutionary algorithm, variation operators act on solutions whose representation is known in advance. In MPaGE and BMAB-LLM, the variation operators are prompt templates applied to source code. Mutation may ask the LLM to improve or diversify one parent heuristic, while crossover may ask it to combine ideas from two parent heuristics. The offspring is therefore a program, not a vector. Evaluation is also more expensive and more failure-prone, because the generated program must first satisfy the task interface before it can be scored.

The use of a Pareto population at the outer level remains essential. If the system selected only the single heuristic with the largest inner hypervolume, it could discard faster heuristics that are practically useful or semantically diverse heuristics that later serve as good parents. Maintaining a front of heuristic programs gives the LLM richer context for future generation and reduces the risk of collapsing to one local programming pattern.

2.5 Hypervolume as a Quality Indicator

The hypervolume indicator measures the volume of the region dominated by a set of objective vectors and bounded by a reference point [5, 6]. For minimization, larger hypervolume generally indicates that the approximation set is closer to the Pareto-optimal region and covers a broader range of trade-offs. Hypervolume is widely used because it combines convergence and diversity into a single scalar indicator, although its computational cost grows with the number of objectives.

The project uses hypervolume at two different levels. The *inner hypervolume* is computed over solution archives produced by a generated heuristic on a benchmark problem. This value measures how well a heuristic solves a task such as Bi-TSP or Bi-CVRP. The *outer hypervolume* is computed over the population of generated heuristics, where each heuristic is represented by a score vector of the form $(-HV_{\text{inner}}, runtime)$. The first coordinate is negated so that the outer population manager can minimize both coordinates.

This distinction is essential when interpreting the experimental results. The figures

and tables in Chapter 5 report heuristic-population hypervolume and AUBC under the `mpage_bmab` harness. They should not be confused with the task-level solution hypervolume tables reported in the original MPaGE paper.

The two hypervolume levels have different reference points because they live in different spaces. For an inner Bi-TSP archive, the points are tour-cost vectors. For an inner Tri-TSP archive, the points are three-dimensional tour-cost vectors. For the outer population, however, every heuristic is represented by a two-dimensional score vector $(-HV_{\text{inner}}, runtime)$, even when the inner task is tri-objective. Thus, the outer hypervolume reference point is a bound in heuristic-score space, not in solution-objective space. The project documentation `mpage_bmab/documents/HV_TWO_LEVELS.md` emphasizes this distinction because confusing the two spaces can lead to incorrect AUBC computation and misleading comparisons.

The outer hypervolume is also sensitive to the range of the first score component. For tasks where inner hypervolume values are large, such as Tri-TSP, the dominated extent along the negated-inner-HV axis can be much larger than in Bi-TSP. Consequently, absolute outer-HV values should not be compared directly across tasks. The thesis therefore uses within-cell normalized scores and ranks when summarizing performance across task families. Absolute values remain meaningful within a task–budget cell, where all methods are evaluated under the same task, budget, reference point, and aggregation convention.

In the implementation, hypervolume computation is centralized in `mpage_bmab/reward.py`. When `pymoo` is available, the project uses `pymoo.indicators.hv.HV` [7]; otherwise, it falls back to an internal two-dimensional implementation for supported cases.

2.6 MPaGE

MPaGE, Pareto-Grid-Guided Large Language Models, is the direct methodological baseline for this project [1]. It frames heuristic design as an LLM-guided program search problem for multi-objective combinatorial optimization. Its workflow combines generated Python heuristics, execution-based evaluation, population management, Pareto Front Grid selection, and semantic clustering.

The first key component is a SEMO-style heuristic population. Each generated heuristic is evaluated and inserted into a population according to multi-objective criteria. The population favors non-dominated heuristics and uses diversity mechanisms to avoid collapsing to a narrow set of similar programs. The population code inherited from MPaGE is available under `llm4ad/method/LLMPFG/population.py`.

The second key component is Pareto Front Grid (PFG) selection. PFG discretizes the heuristic objective space and selects representatives from different grid regions. This

mechanism is designed to preserve useful variation along the heuristic Pareto front. The parameters `pfg_gk`, `pfg_sigma`, and `pfg_epsilon` are documented in `mpage_bmab/documents/HYPERPARAMETERS.md` and retained in the BMAB implementation.

The third key component is LLM-based semantic clustering. Rather than grouping heuristics only by numerical score, MPaGE asks an LLM to cluster heuristic descriptions or code by algorithmic similarity. Mutation is then applied within a semantic cluster, while crossover can combine heuristics across clusters. The relevant prompt families are defined in `llm4ad/method/LLMPFG/prompt.py`.

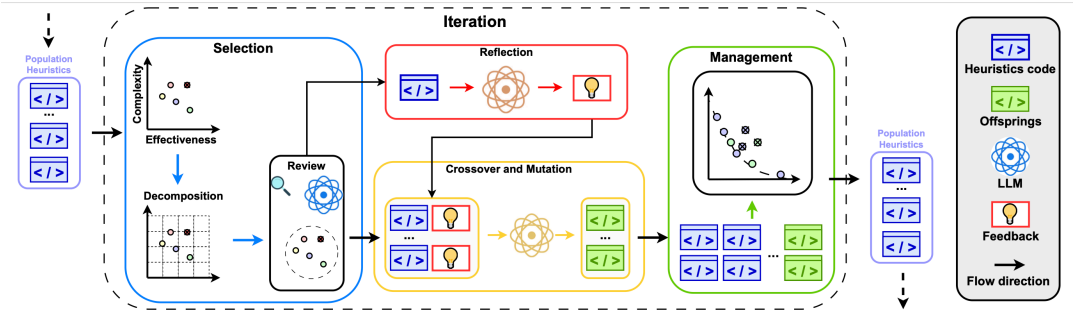


Figure 2.1: Overview of the MPaGE framework. Source figure: `figure/Overview.png`.

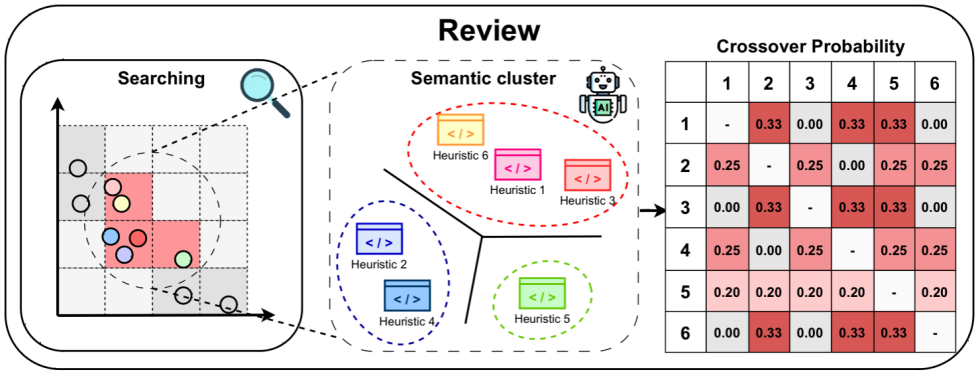


Figure 2.2: Pareto Front Grid selection in MPaGE. Source figure: `figure/Selection.png`.

The proposed BMAB-LLM method does not replace these components. Instead, it keeps the MPaGE generation and evaluation substrate and modifies the scheduling policy that decides how to spend LLM calls.

2.7 LLM-Based Program Generation for Heuristic Design

The project belongs to the broader area of LLM-based program synthesis with execution feedback. In systems such as FunSearch, an LLM generates candidate programs, an evaluator scores them, and the best programs are retained for further search [8]. MPaGE and

BMAB-LLM apply the same general principle to heuristic design: generated code is not accepted because it is plausible natural language output, but because it passes execution and improves a measurable optimization criterion.

In this repository, the LLM is used primarily for prompt-driven code generation and semantic clustering. The prompt templates request Python functions with a specified signature. The secure evaluation layer then executes the generated function under task-specific constraints. This design creates a closed loop: generation proposes code, evaluation supplies a score, and population management determines whether the generated heuristic should influence future prompts.

The repository does not contain evidence of LLM fine-tuning, supervised training, LoRA, PEFT, knowledge distillation, retrieval-augmented generation, diffusion models, or GAN-based generation. The generative component is therefore best described as prompt-based program generation using an external LLM API.

Execution feedback is the key mechanism that makes prompt-based generation usable for optimization. The LLM is not assumed to know whether a generated heuristic is correct or effective. Instead, correctness and usefulness are determined by running the generated code. This makes the evaluator a central part of the learning loop: it converts an untrusted program string into either a numerical score or a failure signal. The approach is similar in spirit to program-search systems such as FunSearch [8], where the model proposes code and the environment supplies objective feedback.

The downside is that execution feedback is expensive and noisy. A generated function may look plausible in natural language but fail at runtime because it uses the wrong variable name, returns an infeasible object, or violates a hidden representation constraint. Even valid code may be behaviorally redundant, producing a neighbor operator that is nearly identical to an existing heuristic. The proposed BMAB controller is designed for precisely this setting: rewards are sparse, failures are common enough to matter, and exploration must be controlled under a finite call budget.

2.8 Validity, Null Scores, and Program-Search Failure Modes

In this repository, an invalid heuristic is not merely a low-quality heuristic. It may fail before producing any meaningful score. The evaluation infrastructure can return a null score when parsing, execution, timeout handling, or task-specific validation fails. These null outcomes are important because they consume budget without directly improving the heuristic population.

There are several qualitatively different failure modes. A syntactic failure means the

LLM output cannot be parsed as executable Python. An interface failure means the function exists but does not match the evaluator's expected signature or return type. A semantic failure means the function runs but returns an infeasible neighbor, such as an invalid tour, overloaded route, or capacity-violating knapsack vector. A runtime failure means the function exceeds the allowed evaluation time or raises an exception during repeated inner-loop calls. The current result summaries do not separate all of these failure types; they aggregate them through valid-yield and invalid/null diagnostics. This is sufficient for the thesis comparisons, but finer-grained failure logging would be valuable future work.

Validity matters differently from final quality. A method that produces many valid heuristics has more opportunities to improve the outer population, but validity alone does not guarantee high hypervolume. A valid heuristic can still be dominated, redundant, or too slow. Therefore, Chapter 5 treats valid heuristic yield as a diagnostic metric rather than a primary performance objective. The central empirical question is whether budget-aware scheduling produces valid programs that also contribute useful objective-space trade-offs.

2.9 Adaptive Operator Selection

Adaptive operator selection studies how an evolutionary algorithm can choose among variation operators based on their observed usefulness during the run. Early work shows that reward-based operator control can improve search by allocating more trials to operators that produce valuable offspring [9]. This idea is directly relevant to LLM-based heuristic design because mutation and crossover prompts do not have uniform value across all stages of the search.

The difference is the cost model. In a conventional evolutionary algorithm, applying a mutation operator may be computationally cheap. In this project, applying a mutation or crossover prompt consumes an LLM call. As a result, operator selection is not merely a performance-tuning mechanism; it is a budget-allocation policy. This observation motivates the operator-level UCB component in `mpage_bmab/bandit.py`.

A second difference is semantic locality. A mutation prompt is usually expected to exploit a single parent by changing or improving its algorithmic idea. A crossover prompt combines two parents and may explore a larger region of program space. Neither operator is uniformly preferable. Mutation can be effective when a cluster already contains strong heuristics whose local variants are likely to remain valid. Crossover can be useful when the current population contains complementary ideas in different semantic clusters. The operator controller therefore has to learn from the observed reward stream rather than relying on a fixed schedule.

2.10 Multi-Armed Bandits and UCB

A multi-armed bandit models sequential decision-making under uncertainty. At each step, the algorithm selects an arm and observes a reward. It must balance exploitation of arms with high empirical reward and exploration of arms whose reward estimates remain uncertain. UCB1 formalizes this trade-off with an optimism bonus [10, 11]:

$$\mathrm{UCB}(a) = \hat{\mu}_a + c\sqrt{\frac{\log t}{n_a}}, \tag{2.6}$$

where $\hat{\mu}_a$ is the empirical mean reward of arm a, n_a is the number of times arm a has been selected, t is the total number of selections, and c controls exploration.

BMAB-LLM uses this principle at two levels. The operator bandit persists across generations and chooses between mutation and crossover. The cluster bandit is reconstructed within a generation and chooses semantic cluster–operator arms. This two-level structure reflects the fact that operator identities are stable across the entire run, whereas cluster identities are local to a particular clustering call.

The bandit formulation is an approximation, not a claim that the heuristic-design environment is a perfectly stationary stochastic bandit. Rewards depend on the current population, the LLM response, the evaluator, and the remaining budget. Nevertheless, the bandit abstraction is useful because it provides an explicit mechanism for balancing empirical success and uncertainty. It also gives a disciplined alternative to ad hoc scheduling rules such as always alternating mutation and crossover or selecting clusters uniformly.

In the thesis setting, an arm pull corresponds to spending at least one LLM call. The reward is computed after the generated child is evaluated. This delayed feedback is still short enough to be used online: after every candidate heuristic, the operator and cluster statistics can be updated before choosing the next candidate. Thus, the method creates a closed loop between generation, evaluation, and future call allocation.

2.11 Budgeted Bandits

Budgeted bandits extend the standard bandit setting by assigning costs to actions and requiring the policy to operate under a finite resource constraint [12]. This is a natural model for LLM-based search, where each call to the generator or clusterer consumes budget. A method that ignores cost may obtain high reward only by using many calls inefficiently, which is undesirable in practical settings.

The implementation in `mpage_bmab/budget.py` exposes this constraint through a hard `BudgetTracker`. The bandit implementation in `mpage_bmab/bandit.py` uses reward-per-cost at the operator level and a budget-aware exploration term at the cluster level.

This is not claimed to be an optimal solution to all budgeted bandit formulations. Rather, it is a pragmatic scheduling mechanism that is compatible with the MPaGE loop and with the project objective of improving hypervolume progress per LLM call.

Formally, if action a_t at decision time t produces reward r_t and consumes cost c_t, the budgeted objective can be written as

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=1}^{T} r_t \right] \quad \text{subject to} \quad \sum_{t=1}^{T} c_t \leq B, \tag{2.7}$$

where π is the scheduling policy and B is the available call budget. In the current implementation, the most important costs are clustering calls and heuristic-generation calls. Both are charged explicitly. The hard budget tracker ensures that the method cannot silently exceed the configured budget, which is essential for fair comparison with MPaGE-orig.

The remaining-budget term in the cluster bandit has a practical interpretation. Early in a run, exploration is valuable because the controller has little evidence about which semantic regions are useful. Near the end of a run, exploration becomes riskier because a failed exploratory call cannot be compensated by many later calls. Scaling the exploration bonus by the remaining budget fraction therefore shifts the policy from exploratory to exploitative behavior as the horizon approaches. This does not make the policy optimal, but it matches the operational goal of protecting final hypervolume while still allowing early discovery.

2.12 Page–Hinkley Drift Detection

The reward distribution in heuristic design is non-stationary. A semantic cluster may initially contain promising parents, but after repeated exploitation its marginal value can decrease. Conversely, a previously weak cluster may become useful after the population changes. Standard UCB estimates can become overconfident when the reward distribution changes.

Page–Hinkley is an online change-detection test for shifts in the mean of a sequential signal [13]. In `mpage_bmab/bandit.py`, `PageHinkleyState` tracks a running mean and a cumulative deviation statistic. When the gap between the historical maximum and the current cumulative statistic exceeds a threshold, the corresponding cluster arm is treated as stale and its statistics are reset. The default parameters, $\delta = 0.005$ and threshold 0.5, are documented in `mpage_bmab/documents/HYPERPARAMETERS.md`.

2.13 AUBC: Area Under the Budget Curve

For a budget-constrained search method, the final value alone does not capture how efficiently the method uses its budget. AUBC addresses this issue by integrating hypervolume

over consumed budget:

$$\text{AUBC} = \frac{1}{B} \int_0^B HV(b)\, db, \tag{2.8}$$

where $HV(b)$ is the outer heuristic-population hypervolume after consuming budget b. The profiler implementation in `mpage_bmab/profiler.py` records budget-HV points, anchors the curve at $(0, 0)$, and extends the last observed value to the total budget when necessary. The metric is documented in `mpage_bmab/documents/METRICS_AUBC_HV.md`.

AUBC is particularly appropriate for this project because the goal is not only to find a strong final heuristic population, but also to spend LLM calls productively throughout the run. A method with higher AUBC can be preferable even when its final hypervolume is similar, especially in interactive or cost-sensitive settings where early stopping is possible.

The normalization by B makes AUBC interpretable as an average hypervolume value over the budget horizon. If two methods end with the same $HV(B)$, the method with the higher AUBC reached useful populations earlier. If one method has higher final HV but lower AUBC, the result indicates a trade-off between early progress and terminal quality. Chapter 5 uses exactly this distinction: final HV identifies the quality of the final managed population, while AUBC identifies how much value was obtained over the entire budget trajectory.

The profiler anchors the budget curve at $(0, 0)$ before the first recorded point. This convention reflects the fact that before any budget is consumed, no generated heuristic population exists under the run. The profiler also extends the last observed hypervolume to the total budget when the final recorded point does not exactly equal B. This prevents underestimating methods whose last population update occurs slightly before the budget is exhausted.

2.14 Technique Inventory

Table 2.1 summarizes the generative AI and machine learning techniques that are either present or absent in the repository. The table is included to keep the scope of the thesis precise: only techniques with direct evidence in the codebase or documentation are analyzed as part of the project.

Table 2.1: Inventory of generative AI and machine learning techniques in the project.

Technique	Status	Evidence in the repository
Large language models	Present	`README.md`, `mpage_bmab/README.md`, `mpage_bmab/main.py`, `mpage_bmab/cluster_manager.py`.
Prompt engineering	Present	I1, E1, E2, M1, M2 generation prompts and the semantic-clustering prompt in `llm4ad/method/LLMPFG/prompt.py`.
Bandits / lightweight reinforcement learning	Present	UCB1, Budgeted UCB, and Page–Hinkley logic in `mpage_bmab/bandit.py`.
Execution-based evaluation of LLM-generated code	Present	`llm4ad/base/evaluate.py` and the evaluators under `mpage_bmab/_llm4ad/task/optimization/`.
RAG	Not found	No information was found in the project source code or documentation.
Fine-tuning, PEFT, LoRA	Not found	No information was found in the project source code or documentation.
Diffusion models, GANs	Not found	No information was found in the project source code or documentation.
Knowledge distillation	Not found	No information was found in the project source code or documentation.
Multi-agent system	Not found	The system contains an LLM–evolution loop and two bandit controllers, but no evidence was found for multiple autonomous agents.

2.15 Limitations of Existing Approaches

The baseline MPaGE framework provides a strong foundation for LLM-based multi-objective heuristic design, but it does not explicitly optimize the allocation of LLM calls under a hard budget. Its semantic clustering and PFG selection maintain diversity, yet they do not by themselves answer which operator and which semantic region should receive the next call. This leaves room for waste when some clusters produce invalid code, redundant heuristics, or low marginal hypervolume improvement.

BMAB-LLM addresses this gap by treating generation as a sequential budgeted decision problem. The method is still limited by the stochasticity of external LLM APIs, the

possibility of invalid generated code, evaluator runtime costs, and the small number of available experimental seeds. These limitations motivate both the implementation safeguards discussed in Chapter 4 and the empirical caution applied in Chapter 5.

Chapter 3

Proposed Method

3.1 Research Motivation and Design Rationale

The proposed method, denoted BMAB-LLM, is designed as a budget-aware extension of the original MPaGE framework. The central observation behind the project is that LLM-driven heuristic design is not only an optimization problem over generated programs, but also a resource-allocation problem. Each heuristic-generation, clustering, or optional reflection step consumes an LLM call, and therefore consumes time, money, and limited experimental budget. A method that produces high-quality heuristics only after many unproductive calls is less useful than a method that converts early calls into a strong heuristic Pareto front.

The original MPaGE paper formulates automatic heuristic design for multi-objective combinatorial optimization problems (MOCOPs) as a Language Multi-Criteria Heuristic Design problem. It combines three important ingredients: a SEMO-style heuristic population, Pareto-Front-Grid (PFG) elite selection, and LLM-based semantic clustering of heuristic code [1, 4]. This design is well motivated because it searches for a population of heuristics that trade off solution quality and runtime, rather than a single best heuristic. It also recognizes that code-level diversity is not equivalent to semantic diversity: two heuristics may be syntactically different but implement nearly identical search logic.

However, the baseline MPaGE workflow allocates generation effort in a mostly predetermined manner. Each generation repeatedly selects elites with PFG, groups promising heuristics into semantic clusters, applies mutation or crossover prompts, evaluates the generated program, and updates the population. This loop is effective for maintaining semantic and objective-space diversity, but it does not explicitly learn which operator or which semantic region is producing the most useful heuristics under the current budget. The stopping condition is also naturally expressed in terms of iterations or sample counts, whereas practical LLM-based search is constrained by a finite number of API calls. Under a strict budget B, the system should therefore learn where to spend the remaining LLM calls instead

of treating all operators and clusters as equally promising.

BMAB-LLM therefore keeps the productive heuristic-generation mechanisms of MPaGE and replaces only the scheduling layer. In particular, the project preserves the original task templates, prompt taxonomy, secure evaluator, PFG-based population management, and LLM semantic clustering implementation. The new contribution is a two-level budgeted multi-armed bandit controller that decides which variation operator and which semantic cluster should receive each LLM call. This design follows a conservative research principle: the comparison against MPaGE is made meaningful by reusing the same generation and evaluation substrate, while changing the decision policy that allocates budget.

3.2 Problem Formulation

3.2.1 Inner Multi-Objective Combinatorial Optimization Problem

Let $\mathcal{X}$ be a finite or discrete feasible set for a MOCOP, and let

$$f(x) = (f_1(x), f_2(x), \ldots, f_M(x)), \qquad x \in \mathcal{X}, \tag{3.1}$$

where all objectives are minimized. For two feasible solutions $x_a, x_b \in \mathcal{X}$, x_a Pareto-dominates x_b, written $x_a \prec x_b$, if

$$f_i(x_a) \leq f_i(x_b) \quad \forall i \in \{1, \ldots, M\}, \qquad \exists j : f_j(x_a) < f_j(x_b). \tag{3.2}$$

The Pareto front is the image of the non-dominated solution set in objective space. Its quality is commonly summarized by the hypervolume indicator $HV_r(\cdot)$ with respect to a reference point r [5, 6]. For a finite objective set Y in a minimization problem, the hypervolume is the Lebesgue measure of the region dominated by the non-dominated points in Y and bounded by r:

$$HV_r(Y) = \lambda\left(\bigcup_{y \in \mathrm{ND}(Y)} [y_1, r_1] \times \cdots \times [y_M, r_M]\right),$$

where $\mathrm{ND}(Y)$ denotes the non-dominated subset and $\lambda(\cdot)$ denotes M-dimensional volume. A larger hypervolume means that the obtained Pareto approximation is both closer to the ideal region and more broadly spread across trade-offs. The benchmark evaluators in this project use this quantity as the inner solution-quality signal and execution time as the efficiency signal.

3.2.2 Outer Heuristic-Design Problem

Following the LMHD formulation in the original MPaGE paper, a heuristic $h \in \mathcal{H}$ is treated as a program that maps the current SEMO archive to a new candidate solution:

$$h : 2^{\mathcal{X}} \to \mathcal{X}, \qquad x' = h(A), \tag{3.3}$$

where A is the current archive of non-dominated solutions maintained by the inner solver. The evaluator assigns each generated heuristic a vector-valued score

$$E(h) = (e_1(h), e_2(h)) = (-HV_{\text{inner}}(h), T(h)) \,. \tag{3.4}$$

Equation 3.4 should be read as the interface between heuristic design and the inner solver. The LLM does not directly output a TSP tour, CVRP route, or knapsack solution. Instead, it outputs a heuristic program h. During evaluation, the inner SEMO-style solver repeatedly calls h with its current archive A and expects one candidate solution x'. The quality of h is therefore determined indirectly by the Pareto archive that the inner solver obtains after running with that heuristic.

Here $HV_{\text{inner}}(h)$ is the hypervolume achieved by the solutions produced when h is used inside the benchmark solver, and $T(h)$ is the measured runtime. More concretely, if the inner run using h returns a non-dominated solution archive A_h, then $HV_{\text{inner}}(h)$ is computed from the objective vectors $\{f(x) : x \in A_h\}$ using the task-specific reference point. This is the *inner hypervolume*: it measures how good one generated heuristic is at solving the underlying combinatorial optimization task.

The first coordinate in $E(h)$ is negated because the outer population-management code is written as a minimization procedure. Thus, a heuristic with a larger inner hypervolume receives a smaller first score coordinate. The second coordinate is runtime, so smaller is also better. This score convention is implemented by the benchmark evaluators in the vendored MPaGE task modules.

At the outer level, the algorithm maintains a population $\mathcal{P}_t \subset \mathcal{H}$ of valid heuristics and aims to approximate the non-dominated set in the heuristic score space. Let $F_t = \{E(h) : h \in \mathcal{P}_t\}$ be the outer heuristic front at step t. Given an LLM-call budget B, BMAB-LLM is optimized for two related criteria:

$$\max \; HV_{\text{outer}}(B) = \max \; HV_r(F_B), \tag{3.5}$$

$$\max \; \text{AUBC} = \max \; \frac{1}{B} \int_0^B HV_{\text{outer}}(b) \, db, \tag{3.6}$$

subject to

$$\sum_{t=1}^{T} c_t \leq B. \tag{3.7}$$

The quantity HV_{outer} is the *outer hypervolume*: it is computed over heuristic score vectors rather than over task solutions. A high outer hypervolume means that the final population contains useful trade-offs between inner solution quality and runtime, for example heuristics that achieve high inner HV quickly and heuristics that are slower but stronger. The final HV objective measures terminal population quality, while AUBC measures how rapidly the method converts budget into Pareto-front quality over the whole run. In the current experiments, the default cost model is call-based, i.e. one LLM request consumes one budget unit. The budget tracker also supports a token-labeled accounting mode, but the implemented main loop and reported experiments use fixed call costs.

3.3 System Overview

BMAB-LLM is organized as an adaptive orchestration layer around the MPaGE primitives. Figure 3.1 summarizes the data and control flow.

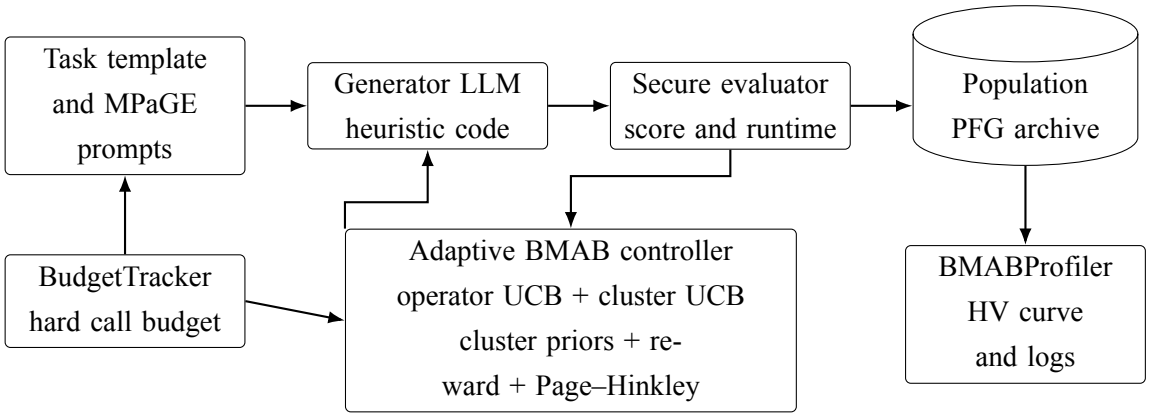


Figure 3.1: Logical architecture of BMAB-LLM. The method preserves MPaGE's prompt, evaluation, semantic clustering, and population-management components, while adding budget accounting and adaptive bandit control.

The implementation is centered in `bmab_llm.py`. The main supporting modules are summarized in Table 3.1.

Table 3.1: Main methodological modules in the implementation.

Module	Methodological role
`bmab_llm.py`	Main orchestrator. It owns the evolutionary loop, calls PFG selection, requests semantic clusters, selects bandit arms, samples offspring, computes rewards, and flushes the final population.
`budget.py`	Hard budget tracker. It stores total, remaining, and consumed budget and records a history entry for every charged LLM call.
`bandit.py`	Two-level adaptive scheduler: persistent operator UCB, per-generation cluster UCB, and Page–Hinkley drift detection.
`cluster_manager.py`	Wrapper around MPaGE semantic clustering. It charges cluster calls, validates or repairs partitions, and computes cluster-quality priors.
`reward.py`	Scalar credit assignment for bandit learning, including HVI, final managed-population HV gain, diversity gain, rank smoothing, and invalid-code penalties.
`profiler.py`	Records budget-vs-HV curves, AUBC, budget history, bandit state, and population artifacts for analysis.

3.4 Components Inherited from MPaGE

BMAB-LLM intentionally retains the core generative and evaluative substrate of MPaGE. This inheritance is documented in the project design notes and is visible in the vendored MPaGE subtree. The retained components are important because they isolate the proposed contribution to budget-aware scheduling rather than changing the benchmark, prompt templates, or evaluation logic.

3.4.1 Population Representation and PFG Selection

Each individual is represented as a Python function together with a natural-language algorithm description, evaluation score, sample time, and execution time. The population class inherited from MPaGE maintains a managed set of heuristics using non-dominated sorting and diversity-preserving truncation.

PFG selection partitions the two-dimensional heuristic objective space into grid cells.

Let $P_t = \{E(h_i) : h_i \in \mathcal{P}_t\}$ be the current score set. For each objective $j \in \{1, 2\}$, define the ideal and nadir values

$$z_j^* = \min_{h \in \mathcal{P}_t} e_j(h), \qquad z_j^n = \max_{h \in \mathcal{P}_t} e_j(h). \tag{3.8}$$

With K_j segments and a small margin σ, the grid width is approximately

$$\Delta_j = \frac{z_j^n - z_j^* + \sigma}{K_j}. \tag{3.9}$$

A heuristic is assigned to a grid index

$$g_j(h) = \left\lfloor \frac{e_j(h) - z_j^*}{\Delta_j} \right\rfloor, \qquad G(h) = (g_1(h), g_2(h)). \tag{3.10}$$

Within each occupied cell, non-dominated representatives are retained. The union of retained representatives forms the elite set used for reproduction. This mechanism is inherited from MPaGE and is implemented in the vendored MPaGE population manager. In the current BMAB loop, the PFG-selected slice is used as a high-quality signal for parent weighting, while the wider managed population is made available to the clustering layer to avoid over-narrow clusters.

3.4.2 LLM-Based Semantic Clustering

The original MPaGE paper argues that objective-space or syntax-only diversity is insufficient for heuristic design. Two code snippets can have different syntax but the same algorithmic logic, and two heuristics with similar performance may use distinct search strategies. MPaGE addresses this by prompting an LLM to group elite heuristics according to semantic and behavioral similarity.

Formally, for a candidate set $P = \{h_1, \ldots, h_n\}$, the clustering module returns

$$\text{SemClust}(P) = \{C_1, \ldots, C_K\}, \tag{3.11}$$

such that

$$\bigcup_{k=1}^{K} C_k = P, \qquad C_i \cap C_j = \emptyset \ (i \neq j). \tag{3.12}$$

The implementation calls the MPaGE cluster prompt through `cluster_manager.py`. If the LLM response is malformed, incomplete, or unavailable, the method falls back to singleton clusters. This design preserves semantic clustering when possible while ensuring that the evolutionary loop remains robust to LLM-format failures.

3.4.3 Mutation and Crossover Prompt Families

MPaGE defines a prompt taxonomy for initialization, mutation, and crossover. BMAB-LLM keeps this taxonomy. Initialization uses the I1 prompt to create zero-shot heuristics. Mutation uses M1 or M2 to modify a single parent from the selected cluster. Crossover uses E1 or E2 to combine one parent from the selected cluster with a second parent outside that cluster. Thus, BMAB preserves MPaGE's core semantic rule:

$$\text{mutation} = \text{intra-cluster local refinement}, \qquad \text{crossover} = \text{inter-cluster recombination}. \tag{3.13}$$

The difference is not the prompt content but the policy that decides when a mutation or crossover prompt should be used, and which cluster should provide the parent.

3.5 Hard Budget Model

The LLM-call budget is treated as a first-class constraint. Let B be the total budget and b_t be the remaining budget before action t. Each LLM action has cost c_t, and the transition is

$$b_{t+1} = b_t - c_t, \qquad b_t \geq c_t. \tag{3.14}$$

If the system cannot afford an action, the action is not issued. The current implementation charges budget before the heuristic-generation call or cluster call is sent. This is important experimentally: invalid code, parsing failure, timeout, and non-improving offspring are still charged because the API resource has already been consumed.

In the default call-based mode, the following costs are used:

Table 3.2: Default budget costs in the current implementation.

Action	Cost
Initialization call (I1)	1
Heuristic-generation call (M1/M2/E1/E2)	1
Semantic clustering call	1
Optional review/suggestion call	1

The budget state is recorded through `BudgetTracker.history`, and the profiler later writes `budget_history.json`. This gives the experimental harness a direct account of how the budget was spent.

3.6 Two-Level Budgeted Bandit Scheduler

3.6.1 Why a Two-Level Scheduler is Needed

A direct bandit over individual heuristics is not viable because the heuristic space is open-ended: every LLM call can create a previously unseen program. A direct persistent bandit over semantic cluster IDs is also problematic because clusters are reconstructed at each generation and their numeric labels are not stable across generations. BMAB-LLM therefore separates the adaptive decision into two levels:

1. a persistent operator-level bandit over the stable action set $\mathcal{O} = \{\mu, \chi\}$, where μ denotes mutation and χ denotes crossover;
2. a per-generation cluster-level bandit over arms (k, o), where k indexes one semantic cluster in the current generation and $o \in \mathcal{O}$ is the selected operator.

This decomposition lets the method learn long-term operator utility while still adapting to the short-lived semantic cluster structure of the current population.

3.6.2 Operator-Level UCB

For each operator $o \in \mathcal{O}$, the operator bandit stores the number of pulls n_o, cumulative reward S_o^R, and cumulative cost S_o^c. Its empirical utility is reward per cost:

$$\hat{\rho}_o = \frac{S_o^R}{\max(S_o^c, \varepsilon)}. \tag{3.15}$$

The selected operator maximizes a UCB score:

$$U_{\text{op}}(o) = \hat{\rho}_o + c_{\text{op}} \sqrt{\frac{2 \log N_{\text{op}}}{n_o}}, \tag{3.16}$$

where $N_{\text{op}} = \sum_{o' \in \mathcal{O}} n_{o'}$ and c_{op} controls exploration. The implementation initializes each operator with an optimistic prior count and prior reward, preventing division by zero and encouraging early exploration. This is implemented in `bandit.py` by the `OperatorBandit` class.

3.6.3 Cluster-Level Budgeted UCB

At the beginning of each generation, `ClusterManager` returns a partition $\{C_1, \ldots, C_K\}$ and a quality prior q_k for each cluster. At this point, q_k can be understood as a scalar estimate of how promising cluster C_k is before any new offspring are generated in the current

generation; its formal front-aware definition is given in Section 3.7. The cluster bandit then creates arms (k, o) for all clusters and operator labels. Each arm stores cumulative reward and cumulative cost, and its exploitation term is

$$\hat{\rho}_{k,o} = \frac{S^R_{k,o}}{\max(S^c_{k,o}, \varepsilon)}. \tag{3.17}$$

The cluster-level UCB score is

$$U_{\text{cl}}(k, o) = \hat{\rho}_{k,o} + c_{\text{cl}}\alpha_t \sqrt{\frac{2 \log N_{\text{cl}}}{n_{k,o}}}, \tag{3.18}$$

where

$$\alpha_t = \left(\frac{b_t}{B}\right)^{\gamma_b} \tag{3.19}$$

when budget annealing is enabled, and $\alpha_t = 1$ otherwise. The factor α_t decreases as the budget is exhausted, making the controller more exploitative near the end of the run. This directly addresses the finite-horizon nature of the task: late calls have little time to recover from risky exploration.

The cluster bandit is warm-started from both cluster quality and operator priors. Let p_o be the softmax-normalized operator utility from the persistent operator bandit. The initial reward assigned to arm (k, o) is proportional to

$$r^{(0)}_{k,o} = \frac{1}{2}\left(q_k + p_o\right). \tag{3.20}$$

This warm start transfers information from the persistent operator layer into the generation-local cluster layer without assuming that cluster IDs persist across generations.

3.7 Front-Aware Cluster Priors

The semantic cluster partition identifies behavioral groups, but it does not by itself rank those groups. The current implementation therefore computes a front-aware prior for each cluster. For cluster C_k, define $S_k = \{E(h) : h \in C_k\}$ and $S = \{E(h) : h \in \mathcal{P}_t\}$. With outer reference point r, the hypervolume contribution of the cluster is

$$\Delta HV_k = \max\left(0,\ HV_r(S) - HV_r(S \setminus S_k)\right). \tag{3.21}$$

The heuristic scores in this section are the two-dimensional evaluator outputs $s = E(h) = (-HV_{\text{inner}}(h), T(h))$. Therefore, s_1 represents negated inner hypervolume and s_2

represents runtime. Because the first score coordinate is negative inner HV, a simple quality proxy for solution strength is

$$I_k = \max\left(0,\ -\min_{s \in S_k} s_1\right), \tag{3.22}$$

with a finite cap in the implementation to avoid an excessively large prior. A runtime proxy is defined as

$$T_k = \frac{1}{1 + \min_{s \in S_k} s_2}. \tag{3.23}$$

The unnormalized prior is

$$\tilde{q}_k = \Delta HV_k + 0.25 I_k + T_k, \tag{3.24}$$

and the normalized prior used by the bandit is

$$q_k = \frac{\tilde{q}_k}{\max_j \tilde{q}_j} \tag{3.25}$$

when the denominator is positive. This design is implemented in `cluster_manager.py`. It is more aligned with the final outer-front objective than a one-dimensional score proxy because it rewards clusters that currently contribute to the heuristic Pareto front while still accounting for inner solution quality and runtime.

3.8 Reward and Credit Assignment

3.8.1 Reward Components

The reward function converts the outcome of one LLM generation call into a scalar signal for the two bandits. For a valid generated heuristic with score s, the reward is

$$R(s) = w_q Q(s) + w_d \max(0, \Delta D) + w_r \text{Rank}(s). \tag{3.26}$$

If the generated heuristic is invalid, cannot be parsed, fails evaluation, or returns a non-finite score, the reward is

$$R(s) = -\lambda_{\text{pen}}. \tag{3.27}$$

The default values in the current implementation are $w_q = 1.0$, $w_d = 0.3$, $w_r = 0.2$, and $\lambda_{\text{pen}} = 1.0$. These settings are exposed through the command-line interface for the weights and fixed in `RewardComputer` for the penalty.

3.8.2 Quality Signal

Let F be the set of valid population scores before the new candidate is inserted. The immediate hypervolume improvement is

$$\mathrm{HVI}(s; F) = \max\left(0,\ HV_r(F \cup \{s\}) - HV_r(F)\right). \tag{3.28}$$

The implementation also computes a managed-population HV delta. Let $\mathrm{Manage}_N(\cdot)$ denote the same score-only survivor-selection logic used by the managed population with cap N. Then

$$\Delta HV_{\text{managed}} = \max\left(0,\ HV_r(\mathrm{Manage}_N(F \cup \{s\})) - HV_r(\mathrm{Manage}_N(F))\right). \tag{3.29}$$

Both quantities are normalized by rolling maxima over a recent window. Let $\widehat{\mathrm{HVI}}$ and $\widehat{\Delta HV}_{\text{managed}}$ denote the normalized values. The quality mode is then

$$Q(s) = \begin{cases} \widehat{\mathrm{HVI}}(s; F), & \text{mode} = \text{dense}, \\ \frac{1}{2}\widehat{\mathrm{HVI}}(s; F) + \frac{1}{2}\widehat{\Delta HV}_{\text{managed}}(s; F), & \text{mode} = \text{hybrid}, \\ \widehat{\Delta HV}_{\text{managed}}(s; F), & \text{mode} = \text{final\_hv}. \end{cases} \tag{3.30}$$

The default `full` method uses `final_hv`. The ablations `dense_reward` and `hybrid_reward` change only this quality signal while retaining the same BMAB pipeline. This design makes it possible to isolate whether terminal managed-population HV, immediate HVI, or a mixture of the two provides the most useful credit signal.

3.8.3 Diversity and Rank Smoothing

Pure HVI is often sparse: many valid offspring do not immediately expand the outer Pareto front. To provide a denser signal, BMAB-LLM adds rank and diversity terms. The cumulative diversity index used in the reward is the mean pairwise Euclidean distance among valid score vectors:

$$D(F) = \frac{1}{|F|(|F|-1)} \sum_{i \neq j} \|s_i - s_j\|_2. \tag{3.31}$$

The diversity increment is $\Delta D = D(F \cup \{s\}) - D(F)$. Only positive increments are rewarded. The rank score is

$$\mathrm{Rank}(s) = \frac{1}{|F|}\left|\{u \in F : \exists j,\ s_j < u_j\}\right|. \tag{3.32}$$

In this equation, s and u are heuristic scores of the form $E(h) = (-HV_{\text{inner}}(h), T(h))$. The first coordinate measures negated inner hypervolume, and the second coordinate measures runtime. Since both coordinates are minimized in the outer heuristic population, a smaller coordinate value indicates either higher inner solution quality or lower execution time. The rank term therefore rewards a candidate that improves at least one objective relative to existing population members, even when it does not immediately increase HV.

3.9 Non-Stationarity and Page–Hinkley Reset

The reward distribution in evolutionary heuristic design is non-stationary. A cluster that is useful early may become unproductive after its design pattern has already been exploited. To reduce overconfidence in stale arm statistics, each cluster-level arm is monitored by a Page–Hinkley detector [13].

For a reward sequence (r_t) of one arm, the implementation maintains the running mean $\bar{r}_t$ and cumulative statistic

$$\bar{r}_t = \bar{r}_{t-1} + \frac{r_t - \bar{r}_{t-1}}{t}, \tag{3.33}$$

$$m_t = m_{t-1} + r_t - \bar{r}_t + \delta, \tag{3.34}$$

$$M_t = \max(M_{t-1}, m_t). \tag{3.35}$$

Downward drift is declared when

$$M_t - m_t > \lambda_{\text{PH}}. \tag{3.36}$$

When drift is detected, the corresponding arm statistics are reset to an optimistic prior for the current generation. This mechanism is implemented in `bandit.py`. It is limited to the cluster-level bandit because cluster rewards are local to the current generation and most sensitive to rapid changes in population state.

3.10 Parent Selection and Variation Workflow

The original MPaGE strategy uses intra-cluster mutation and inter-cluster crossover. BMAB-LLM preserves this logic but changes how parents are sampled. The loop first obtains PFG-selected elites through `Population.selection`. It then expands the clustering pool to the full current managed population to give the cluster bandit a richer set of semantic groups. To prevent the wider pool from diluting the PFG signal, parent sampling gives candidates that also appear in the PFG-selected elite set a larger weight:

$$w(h) = \begin{cases} 3, & h \in E_{\text{PFG}}, \\ 1, & \text{otherwise.} \end{cases} \tag{3.37}$$

For mutation, a single parent is sampled from the selected cluster using these weights. For crossover, the first parent is sampled from the selected cluster and the second parent is sampled from other clusters when possible. This preserves semantic recombination while biasing the process toward strong PFG representatives. In the implementation, this logic is used when constructing the mutation or crossover prompt and when sampling parents from the selected semantic groups.

3.11 Complete Algorithm

Algorithm 3.1 gives the high-level workflow implemented by `BMABLLM.run`. The pseudocode abstracts away exception handling and API-specific details, but preserves the execution order that matters methodologically: initialization, budget-aware clustering, adaptive operator and cluster selection, reward computation, bandit update, drift handling, and final population flushing.

Listing 3.1: High-level BMAB-LLM workflow.

```
Algorithm BMAB-LLM
Input: evaluator E, generator L, clusterer C,
     budget B, population cap N, reference point r
Output: managed heuristic population P

initialize BudgetTracker(B), Population P
initialize OperatorBandit, ClusterBandit, RewardComputer

while initialization target is not reached
     and one generation call is affordable do
   charge one budget unit
   h <- L(I1 initialization prompt)
   score <- E(h)
   if score is valid then
      register h in P
   end if
end while

record the initial budget-HV point
```

```
while budget remains do
   E_pfg <- PFG-selected elites from P
   H_full <- current managed population

   if a clustering call and a generation call are affordable then
      charge one budget unit
      clusters <- C(H_full)
      if clusters are invalid then
         clusters <- singleton fallback clusters
      end if
   else
      clusters <- singleton fallback clusters
   end if

   q <- front-aware cluster priors
   p <- operator priors from OperatorBandit
   reset ClusterBandit using clusters, q, and p

   produced <- 0
   while produced < N
        and one generation call is affordable do
      op <- select mutation or crossover
      arm <- select a cluster arm for op
      parents <- weighted parent sampling from clusters
      prompt <- MPaGE mutation or crossover prompt

      F_before <- scores in P and pending valid offspring
      D_before <- diversity(F_before)

      charge one budget unit
      h_child <- L(prompt)
      score <- E(h_child)
      if score is valid then
         register h_child as pending
      end if

      F_after <- scores in P and pending valid offspring
      D_after <- diversity(F_after)
      reward <- RewardComputer(score, F_before,
                          D_before, D_after)
```

```
      update OperatorBandit(op, reward)
      update ClusterBandit(arm, reward)
      apply Page-Hinkley reset if drift is detected
      produced <- produced + 1
   end while

   merge pending valid offspring into P when needed
   record bandit state and budget-HV point
end while

flush all pending valid offspring into P
write budget history, budget curve, bandit log, and AUBC
```

3.12 Baseline and Ablation Design

The baseline `mpage_orig` is not a new algorithm. It is a wrapper around the original MPaGE driver that records budget consumption and budget-HV curves using the same profiler infrastructure as BMAB-LLM. This is implemented in `mpage_orig.py`. The purpose is to compare allocation policies while keeping the task evaluators and generated artifacts as consistent as possible.

The current CLI defines the following method variants in `main.py`:

Table 3.3: Main method variants relevant to the proposed method.

Variant	Interpretation
`full`	Proposed BMAB method with the `final_hv` reward mode, operator bandit, cluster bandit, Page–Hinkley, budget annealing, and PFG-weighted parent sampling.
`dense_reward`	BMAB pipeline with the immediate-HVI quality signal used for reward ablation.
`hybrid_reward`	BMAB pipeline with a half immediate-HVI and half managed-final-HV quality signal.
`no_budget_anneal`	Same as `full`, but without remaining-budget scaling in the cluster UCB exploration term.
`no_ph`	Disables practical Page–Hinkley resets by using an extremely large threshold.
`no_diversity`	Removes the CDI diversity term by setting its weight to zero.
`op_only`	Disables the cluster bandit and samples clusters uniformly.
`cluster_only`	Disables the operator bandit and uses a round-robin operator sequence.
`mpage_orig`	Budget-instrumented wrapper around the original MPaGE implementation.

These variants are important because they separate the contribution of each methodological component. In particular, `dense_reward` and `hybrid_reward` vary only the reward quality mode, while `no_budget_anneal`, `no_ph`, `no_diversity`, `op_only`, and `cluster_only` isolate specific control mechanisms.

Chapter 4

System Implementation

4.1 Repository Organization

The repository contains two closely related codebases. The first is the original MPaGE source tree, located mainly under `llm4ad/`, together with the original entry points and documentation. The second is the extended project under `mpage_bmab/`, which implements the budget-aware BMAB controller, experiment harness, analysis documents, result files, and the thesis source.

The main implementation modules in `mpage_bmab/` are the following:

- `bmab_llm.py`: the main BMAB orchestration loop. It performs initialization, PFG selection, semantic clustering, bandit-based operator and cluster selection, offspring generation, reward computation, population update, and final artifact writing.
- `bandit.py`: the implementation of the operator-level UCB controller, cluster-level budgeted UCB controller, and Page–Hinkley drift detector.
- `budget.py`: the hard LLM-call budget tracker. It records consumed and remaining budget and rejects unaffordable actions.
- `cluster_manager.py`: the wrapper around LLM-based semantic clustering. It charges cluster calls, validates cluster partitions, computes cluster priors, and provides fallback behavior.
- `reward.py`: the reward computation module, including hypervolume, immediate HVI, managed final-HV gain, diversity gain, rank smoothing, and invalid-code penalties.
- `profiler.py`: the logging and measurement module that records budget curves, AUBC, bandit logs, budget history, and population-level artifacts.

- `mpage_orig.py`: the wrapper that runs the original MPaGE baseline under the same budget-accounting and output conventions used by BMAB.
- `experiments/`: the experiment harness, including suite definitions, run scripts, aggregation, comparison, metric recomputation, diagnostics, and figure-generation code.

This organization separates methodological logic from experiment execution. The core algorithm is implemented in `bmab_llm.py`, while the repeatable experimental matrix is controlled by `experiments/configs.py` and `experiments/run.py`.

The separation is important for thesis reproducibility. Methodological claims in Chapter 3 are tied to the algorithm modules, while empirical claims in Chapter 5 are tied to the experiment and analysis modules. This avoids mixing two different concerns: changing the scheduler should happen in the core BMAB implementation, whereas changing which tasks, budgets, or seeds are run should happen in the experiment harness. The current repository mostly follows this separation, which makes it possible to regenerate result tables without rewriting the method itself.

4.2 Software Environment

The root `requirements.txt` and `mpage_bmab/requirements.txt` list the main Python dependencies. The important libraries include `numpy`, `scipy`, `pymoo`, `matplotlib`, `pandas`, `tqdm`, and an OpenAI-compatible client. The command-line interface in `mpage_bmab/main.py` uses `gpt-4o-mini` as the default generator and clusterer model unless overridden by CLI arguments. API credentials are read from local secret files; their contents are not part of the thesis and are not required for interpreting the algorithmic design.

The available repository materials do not specify the exact hardware, GPU, operating system version, API latency, token cost, or monetary cost per call used to produce the reported result set. No information was found in the project source code or documentation for these environment details. For this reason, the thesis treats budget primarily as the number of charged LLM calls rather than as wall-clock cost or monetary cost.

4.3 Benchmark Evaluators and Task Preparation

The current BMAB implementation imports task evaluators from the project-local vendored tree `mpage_bmab/_llm4ad/task/optimization/`. Each task defines both a template

for the generated heuristic function and an evaluator that executes the generated function inside an archive-based inner search procedure.

Bi-TSP is implemented under `mpage_bmab/_llm4ad/task/optimization/bi_tsp_semo/`. The evaluator uses four fixed instances, problem size 20, a timeout of 60 seconds, an initial archive of 100 random tours, and 2,000 heuristic calls per instance.

Tri-TSP is implemented under `mpage_bmab/_llm4ad/task/optimization/tri_tsp_semo/`. The evaluator uses 20 fixed instances, problem size 20, a timeout of 90 seconds, an initial archive of 100 random tours, and 20,000 heuristic calls per instance.

Bi-CVRP is implemented under `mpage_bmab/_llm4ad/task/optimization/bi_cvrp/`. The evaluator uses eight fixed instances, problem size 100, capacity 50 for this size range, a timeout of 90 seconds, 10 initial route sets, and 6,000 heuristic calls per instance.

Bi-KP is implemented under `mpage_bmab/_llm4ad/task/optimization/bi_kp/`. The evaluator uses eight fixed instances, problem size 200, capacity 25 for this size range, a timeout of 90 seconds, 20 initial feasible item selections, and 8,000 heuristic calls per instance.

The `get_instance.py` files in these task directories set the NumPy random seed to 2025 before generating benchmark instances. Therefore, the problem instances are fixed under the current code. The experiment seeds 2025–2029 mainly affect the outer heuristic-generation and selection process rather than creating new benchmark instance sets.

4.4 Score and Population Data Model

The core data object flowing through the system is a generated heuristic function. Conceptually, a function object stores the generated code, metadata, and an optional score. A score is valid only when the code can be executed by the corresponding task evaluator and returns behavior that can be measured. For all benchmark tasks analyzed in this thesis, the outer score has two components:

$$s(h) = \left(-\overline{HV}_{\text{inner}}(h), \overline{\tau}(h)\right). \tag{4.1}$$

The first component is negated because the outer population manager follows a minimization convention. The second component is average runtime. This two-dimensional convention is used even for Tri-TSP, whose inner archive has three objective dimensions. The tri-objective inner archive is first summarized by inner hypervolume, and that scalar is then paired with runtime at the outer level.

The population manager keeps a bounded set of generated heuristics. New children enter a pending population during a generation and are later merged through survivor se-

lection. This detail is significant under a strict budget. If a run stops in the middle of a generation, pending valid children must still be considered before the final population is recorded. The finalized implementation includes this flush behavior, which prevents useful generated heuristics from being lost simply because the budget ended before a full generation was completed.

Dominance in the outer population is evaluated over heuristic scores, not over source-code similarity. Semantic clustering is used for parent organization and budget allocation, but the final population is determined by numerical objectives. Thus, a semantically novel heuristic that is slow and low-quality can still be dominated, while a semantically similar heuristic can survive if it improves the quality–runtime trade-off.

4.5 Heuristic Generation and Evaluation Pipeline

Each generated heuristic is represented as Python code containing a task-specific `select_neighbor` function. The LLM is prompted using MPaGE prompt templates, the response is parsed into executable code, and the evaluator attempts to run the function in a controlled process. A successful evaluation returns a two-dimensional outer score: negated inner hypervolume and runtime. If the code fails to parse, raises an exception, violates task constraints, times out, or returns an invalid object, the evaluator may return a null score.

The project uses the same conceptual evaluation interface across all benchmark families. For each fixed problem instance, the evaluator creates an initial archive, repeatedly calls the generated `select_neighbor` heuristic, inserts feasible and non-dominated candidates, removes dominated archive members, and computes the inner hypervolume of the final archive. The heuristic score used by the outer BMAB population is then averaged over the task instances.

Tri-TSP illustrates the two-level nature of the evaluation. Although the inner task has three distance objectives, the outer heuristic score remains two-dimensional: $(-HV_{\text{inner}}, runtime)$. This convention is reflected in `mpage_bmab/main.py`, where the outer reference point for Tri-TSP is two-dimensional, `(20.0,60.0)`.

4.6 Execution Flow of a BMAB Run

A single BMAB run proceeds through a fixed sequence of stages. First, the command-line entry point in `mpage_bmab/main.py` resolves the task evaluator, population size, budget, seed, model names, and reward-mode settings. It then constructs the budget tracker, profiler, population manager, cluster manager, bandit controllers, and reward computer.

Second, the method performs an initialization phase. Initial heuristic programs are

generated and evaluated until the initial population is available or the budget is exhausted. These initial candidates provide the first set of scores from which PFG selection and semantic clustering can operate. Initialization is therefore not only a warm-up step; it determines the early structure of the outer search space.

Third, each generation starts by selecting elites and preparing a clustering pool. The implementation retains the MPaGE idea that promising heuristic programs should guide the next prompt context, but the finalized BMAB implementation also expands the clustering pool so that the cluster bandit can reason over a richer set of semantic regions. The cluster manager either obtains an LLM-based partition or falls back to a deterministic singleton-style partition when the budget is too small to justify a clustering call.

Fourth, the inner generation loop repeatedly selects an operator and a cluster arm. The operator-level bandit chooses between mutation and crossover. The cluster-level bandit chooses which semantic region should receive the current call under the selected operator family. Parent heuristics are then sampled, with higher weight assigned to parents that also appear in the PFG-selected elite set. The LLM generates a child program, the evaluator scores it, and the reward computer converts the outcome into a scalar feedback signal.

Finally, the run writes artifacts. The profiler records budget-curve points, AUBC, budget history, bandit logs, and population snapshots. The finalization step flushes pending offspring into the managed population before final HV is recorded. This flow makes the final result dependent on all valid generated work, not only on completed generations.

4.7 Budget Accounting and Call Semantics

The project uses call-based budget accounting. A budget unit corresponds to an LLM call made for generation or clustering. This abstraction is simpler than token-level or monetary accounting, but it has two advantages for the current thesis. First, it is directly observable from the code path: every charged action passes through the budget tracker. Second, it is stable across tasks, which makes the task–budget matrix easier to compare.

The budget tracker enforces two properties. The first is monotonic consumption: once a call is charged, the consumed budget cannot decrease. The second is affordability checking: an action should not be taken if the remaining budget cannot pay for it. The finalized method uses this not only as a safety mechanism but also as part of the algorithm. Near the end of a run, the implementation avoids spending the last available unit on clustering if doing so would leave no budget for generating a heuristic. This design choice reflects the principle that a clustering call has value only if it can guide at least one subsequent generation call.

Call-based budget accounting does not capture every practical cost. It ignores prompt

length, completion length, API latency, and evaluator runtime. The repository does not provide these measurements for the result set. Nevertheless, call budget is an appropriate first-order abstraction for comparing scheduling policies because it measures the scarce generative action that the proposed method controls.

4.8 Experiment Configuration

The main experimental matrix is defined in `mpage_bmab/experiments/configs.py`. The default task set is `bi_tsp`, `tri_tsp`, `bi_cvrp`, and `bi_kp`. The default budgets are $B \in \{25, 50, 100, 200\}$, and the default seeds are 2025, 2026, 2027, 2028, and 2029.

The function `pop_size_for` selects the managed population size according to budget. For the two-objective tasks, the population sizes are 4, 6, 8, and 10 for budgets 25, 50, 100, and 200, respectively. For Tri-TSP, the configuration adds two extra individuals, yielding population sizes 6, 8, 10, and 12. This policy is also documented in `mpage_bmab/documents/HYPERPARAMETERS.md`.

The default bandit parameters are $c_{\text{op}} = 1.0$, $c_{\text{cluster}} = 1.0$, $\gamma_{\text{budget}} = 0.5$, prior count 1, and prior reward 0.5. The default reward weights are $w_q = 1.0$, $w_d = 0.3$, and $w_r = 0.2$, with invalid-code penalty 1.0 and rank window 50. Page–Hinkley uses $\delta = 0.005$ and threshold 0.5. These values are exposed or documented through `mpage_bmab/main.py`, `mpage_bmab/reward.py`, `mpage_bmab/bandit.py`, and `mpage_bmab/documents/HYPERPARAMETERS.md`.

The configuration intentionally keeps population size coupled to budget. At small budgets, a large population would consume most of the budget during initialization and leave too few later decisions for the bandit to learn. At large budgets, a larger population can preserve more heuristic diversity and provide richer parent material. Tri-TSP receives a slightly larger population because the inner problem has three objectives and therefore tends to require more diversity to represent useful trade-offs.

4.9 Experiment Suites and Run Scripts

The experiment harness under `mpage_bmab/experiments/` separates suite definition, execution, aggregation, and comparison.

`experiments/run.py` enumerates cells of the form `(ablation,task,budget,seed)`, creates output directories, and dispatches each cell to either `mpage_bmab/main.py` or `mpage_bmab/mpage_orig.py`. It skips completed cells unless `--force` is specified. `experiments/aggregate.py` scans result directories and creates aggregate CSV summaries. `experiments/compare.py` performs paired comparisons against a selected

baseline for a selected metric.

Several shell launchers are provided for common workflows. `experiments/run_full.sh` launches the standard full BMAB ablation grid and then aggregates and compares results. `experiments/run_mpage_compare.sh` launches the MPaGE-orig versus BMAB comparison. `experiments/run_hv_final_full.sh` launches the reward-mode comparison among `full`, `dense_reward`, and `hybrid_reward`, then runs aggregation and comparisons for both `hv_final` and AUBC.

The result set analyzed in Chapter 5 combines the complete MPaGE comparison matrix with the complete reward-mode matrix. It therefore covers four thesis-facing setups: Final-HV reward, Dense reward, Hybrid reward, and MPaGE-orig. Other component ablations, such as `no_ph`, `no_diversity`, `op_only`, and `cluster_only`, are available in the codebase but are not analyzed as complete matrices in Chapter 5.

4.10 Generated Artifacts

Each run writes a structured set of artifacts that makes post-hoc analysis possible. The most important artifacts are:

- `samples/`: generated heuristic samples, including code and available scores.
- `population/`: final and intermediate heuristic-population records.
- `budget_curve.json`: a sequence of points containing consumed budget, outer hypervolume, and Pareto population size.
- `budget_history.json`: the budget-consumption history.
- `bandit_log.json`: selected operators, selected cluster arms, rewards, and related bandit diagnostics.
- `aubc.json`: the final AUBC value computed from the budget curve.
- `run_log.txt`: execution logs for the run.

The numerical summaries used in Chapter 5 are materialized in thesis-local table files under `mpage_bmab/thesis/tables/`. This makes the reported values self-contained in the thesis while preserving compatibility with the experiment aggregation scripts when new runs are added.

The artifact structure also supports anomaly inspection. For example, if a cell has positive final HV but zero AUBC, the corresponding `budget_curve.json` and `aubc.json` files can be checked directly. This was used during the dense-reward Bi-CVRP $B = 25$,

seed-2029 rerun discussed in Chapter 5. The thesis therefore treats generated artifacts not only as outputs but as audit records that make it possible to validate the aggregation pipeline.

Another important artifact is the saved sample set. Sample files make it possible to inspect the generated code and verify whether a run produced valid heuristics, null-score candidates, or repeated patterns. The current analysis does not perform qualitative code motif analysis, but the presence of these artifacts would support such an extension. This is especially relevant for future work on explaining why particular prompt families or semantic clusters become productive.

4.11 Visualization and Analysis Code

The figures used in Chapter 5 are generated from result CSV and JSON files. The consolidated four-setup overview is produced by `mpage_bmab/experiments/analyze_all_setups_overview.py`, which creates AUBC plots, final-HV plots, budget curves, valid-yield charts, invalid/null diagnostics, pairwise win matrices, and trade-off figures. The rerun-specific Bi-CVRP diagnostic figures are produced by `mpage_bmab/experiments/analyze_rerun_cell_update.py`. The thesis includes local copies of the generated figures under `mpage_bmab/thesis/figures/`.

This separation between data, plotting code, and thesis figures is useful for reproducibility. If new runs are added, the same aggregation and visualization pipeline can be rerun to update the numerical summaries and figures without changing the methodological exposition.

The visualization pipeline uses several levels of aggregation. Per-run artifacts are first converted into row-level summaries containing task, budget, seed, setup, AUBC, final HV, valid count, and diagnostic rates. These rows are then aggregated into task–budget cells. Normalized scores are computed within each cell so that tasks with very different absolute hypervolume scales can be compared fairly. Finally, the analysis scripts generate overview plots, pairwise win matrices, and budget-curve figures. This sequence is important because it prevents large-scale tasks such as Tri-TSP from dominating cross-task conclusions simply because their absolute outer-HV values are larger.

The thesis includes figure copies rather than linking directly to the experiment output directory. This is a practical LaTeX decision: it makes the thesis project self-contained and avoids broken figure paths if experiment folders are archived or moved. The original result files remain the source of truth for numerical regeneration, while the thesis-local copies provide a stable compiled document.

4.12 Reproducibility Considerations

The codebase provides several mechanisms for reproducibility: fixed benchmark-instance generation, explicit experiment seeds, deterministic suite definitions, per-run output directories, logged budget curves, and saved generated samples. These mechanisms make it possible to inspect what happened in each run and to recompute aggregate statistics from stored artifacts.

However, exact bit-level reproducibility is not guaranteed. The system depends on an external LLM API, and the repository does not provide a complete snapshot of the model backend, sampling behavior, or service-side changes over time. The generated code can also be stochastic, because heuristics may call `random` or `numpy.random`. Therefore, the thesis interprets the reported results as reproducible under the stored artifacts and code paths, while recognizing that re-running the same experiment at a later date may produce slightly different generated heuristics.

4.13 Implementation Caveats

Three implementation caveats are important when interpreting the results. First, `mpage_orig` and BMAB do not record budget history with identical granularity. BMAB records detailed budget events, whereas the MPaGE wrapper records a more aggregated history. Therefore, the number of budget-history rows should not be compared as if they were identical event types.

Second, BMAB stores only accepted or scored heuristic samples in several output locations, while MPaGE-orig can retain samples with null scores. Consequently, invalid/null-score rate is a diagnostic indicator rather than a perfectly symmetric storage metric. The analysis in Chapter 5 accounts for this by using both stored samples and budget-history-derived counts.

Third, the runtime timeouts in some benchmark configuration files are not always identical to the timeout values used directly in the Python evaluator classes. When there is such a discrepancy, the thesis prioritizes the evaluator source code because it is the executed path used by the current implementation.

Chapter 5

Experiments and Evaluation

5.1 Experimental Questions

The experiments in this chapter evaluate how the proposed budget-aware MPaGE extension behaves under a finite LLM-call budget. The analysis is based on the consolidated overview of all experimental setups and the corresponding raw per-run artifacts. The final comparison contains four complete thesis-facing setups, four benchmark tasks, four budget levels, and five random seeds, giving a total of 320 runs.

The chapter addresses four experimental questions. First, which setup gives the best overall budget-aware performance across tasks and budgets? Second, does a setup that improves budget efficiency, measured by AUBC, also improve terminal heuristic-population quality, measured by final outer hypervolume? Third, how do the reward-mode variants differ from the official finalized method? Fourth, do validity diagnostics, such as valid heuristic yield and invalid/null-score rate, help explain the observed performance differences?

These questions follow from the project objective, the metric definitions, and the reward-mode descriptions documented with the implementation. The thesis does not infer results from undocumented experiments; all numerical claims in this chapter are taken from the generated result files.

5.2 Experimental Setups

The analyzed matrix contains four setups. Table 5.1 summarizes their purpose and configuration. The official proposed method is Final-HV reward, which is the finalized BMAB implementation using `reward_mode=final_hv`. The two reward-mode variants, Dense reward and Hybrid reward, keep the finalized implementation but change the primary quality signal used by the bandit reward. MPaGE-orig is the budget-wrapped original MPaGE baseline.

Table 5.1: Experimental setups analyzed in the final comparison.

Setup	Purpose	Configuration
Final-HV reward	Official finalized BMAB method	Implementation setup identifier `full`. Uses the two-level bandit scheduler, Page–Hinkley drift handling, budget annealing, diversity/rank reward terms, pending-offspring flushing, and `reward_mode=final_hv`.
Dense reward	Reward-mode ablation	Keeps the finalized implementation but changes only the main quality signal to immediate hypervolume-improvement feedback, i.e., `reward_mode=dense`.
Hybrid reward	Reward-mode ablation	Keeps the finalized implementation but uses a half dense and half final-population HV quality signal, i.e., `reward_mode=hybrid`.
MPaGE-orig	Original MPaGE baseline	Runs the original MPaGE workflow through the project wrapper so that budget curves, AUBC, and final-HV artifacts can be collected under the same measurement protocol.

The Final-HV reward setup is the main proposed method. Its implementation identifier is `full`. It uses the two-level budgeted bandit scheduler, front-aware cluster priors, Page–Hinkley drift handling, final pending-offspring flushing, budget-aware exploration annealing, and the final managed-population HV reward. This setup is the one that should be interpreted as the official contribution of the thesis.

The Dense reward setup isolates the effect of the reward quality signal. It preserves the finalized scheduler and implementation fixes but replaces the final-HV quality signal with immediate HVI-style feedback. It therefore answers a controlled ablation question: if all current implementation details are kept constant, what happens when the bandit is trained on dense local improvement instead of final managed-population improvement?

The Hybrid reward setup averages the dense and final-HV signals. Its purpose is to test whether early local feedback and terminal-population alignment are complementary. In principle, a hybrid signal may improve early budget curves because dense HVI is less sparse, while still keeping some pressure toward the final metric.

The MPaGE-orig setup represents the original MPaGE workflow under the project budget wrapper. It is the most important baseline because it preserves the MPaGE generation and population-management philosophy while lacking the BMAB call-allocation controller.

All four setups are evaluated on Bi-TSP, Tri-TSP, Bi-CVRP, and Bi-KP. The budget levels are $B \in \{25, 50, 100, 200\}$, and the seeds are 2025, 2026, 2027, 2028, and 2029. Thus each setup contributes $4 \times 4 \times 5 = 80$ runs. The result summaries show that the current matrix contains 320 aggregated runs. The experiment harness records AUBC, final outer hypervolume, budget curves, budget histories, valid heuristic counts, and invalid/null diagnostics for post-hoc analysis.

The unit of comparison is the task–budget cell. For each setup, a cell aggregates five seeds. Most tables in this chapter first compute a mean within each cell, then compare setups across the 16 cells formed by four tasks and four budgets. This design avoids overweighting any single task or budget level. It also makes the analysis robust to differences in absolute HV scale across tasks, especially between Tri-TSP and the bi-objective tasks.

5.3 Evaluation Metrics

The primary metric is AUBC, the area under the outer hypervolume-vs-budget curve. It measures how effectively a setup converts consumed LLM-call budget into heuristic-population quality throughout a run. A method can be useful under a strict budget even if its final hypervolume is only slightly different, because reaching a strong population earlier provides more value when calls are expensive or a run may be stopped early.

The secondary metric is `hv_final`. In `mpage_bmab/experiments/aggregate.py`, this value is extracted from the final `hv` entry in `budget_curve.json`. It measures the terminal outer hypervolume of the managed heuristic population. It should not be confused with the inner solution-level hypervolume computed by each benchmark evaluator.

The chapter also reports valid heuristic yield per 100 calls and invalid/null-score rate. These are diagnostic metrics rather than direct optimization objectives. They are useful because LLM-generated code may fail to parse, return infeasible outputs, time out, or receive a null score. However, the invalid/null diagnostic is not perfectly symmetric across all setups: BMAB-style runs mainly store valid or scored samples, whereas `mpage_orig` can retain null-score samples differently. Therefore, validity diagnostics are interpreted cautiously and separately from the primary performance metrics.

For budget-curve figures, the horizontal axis is normalized as `budget_consumed/total_budget`. The plotted curves use the same initial anchor as `mpage_bmab/profiler.py`: the curve begins at $(0, 0)$ before the first recorded hypervolume value. This

convention makes the plotted budget curves consistent with the AUBC calculation.

The normalized scores in this chapter are computed within each task–budget cell. If a setup has value $v_{m,c}$ for metric m in cell c, and the best value in that cell is $v^{\star}_{m,c}$, the normalized score is a percentage relative to the cell best. This transformation is especially important for AUBC and HV because absolute values differ greatly across tasks. Without cell normalization, high-magnitude tasks would dominate the overall average even if the relative performance pattern were similar.

Mean rank is reported alongside normalized score because the two summaries answer different questions. Normalized score measures practical closeness to the best observed value. Mean rank measures ordering stability. A method can have a high normalized score but a weaker rank if it is often slightly below the best method. Conversely, a method can achieve a good rank while having a larger gap in a few cells. The thesis therefore interprets both values together.

5.4 Overall Four-Setup Results

Table 5.2 summarizes the four setups using normalized scores across the 16 task–budget cells. A score of 100 means that the setup is the best method in that specific task–budget cell; the table reports the mean of these normalized cell scores. The performance score is the mean of the AUBC and final-HV normalized scores. Figure 5.1 visualizes the same comparison for AUBC, final HV, and valid yield.

Table 5.2: Overall normalized scores and mean ranks across the 16 task–budget cells. Scores are normalized within each task–budget cell, where 100 denotes the best setup in that cell. Lower mean rank is better.

(a) Performance-oriented metrics

Setup	AUBC sc.	AUBC rk.	Final-HV sc.	Final-HV rk.	Perf. sc.
Final-HV reward	96.2	1.88	97.6	1.88	96.9
MPaGE-orig	76.2	3.94	91.2	3.19	83.7
Dense reward	95.4	2.00	96.1	2.50	95.8
Hybrid reward	96.2	2.19	97.6	2.44	96.9

(b) Validity diagnostic

Setup	Valid sc.	Valid rk.
Final-HV reward	90.7	2.41
MPaGE-orig	84.1	3.16
Dense reward	90.8	2.69
Hybrid reward	94.4	1.75

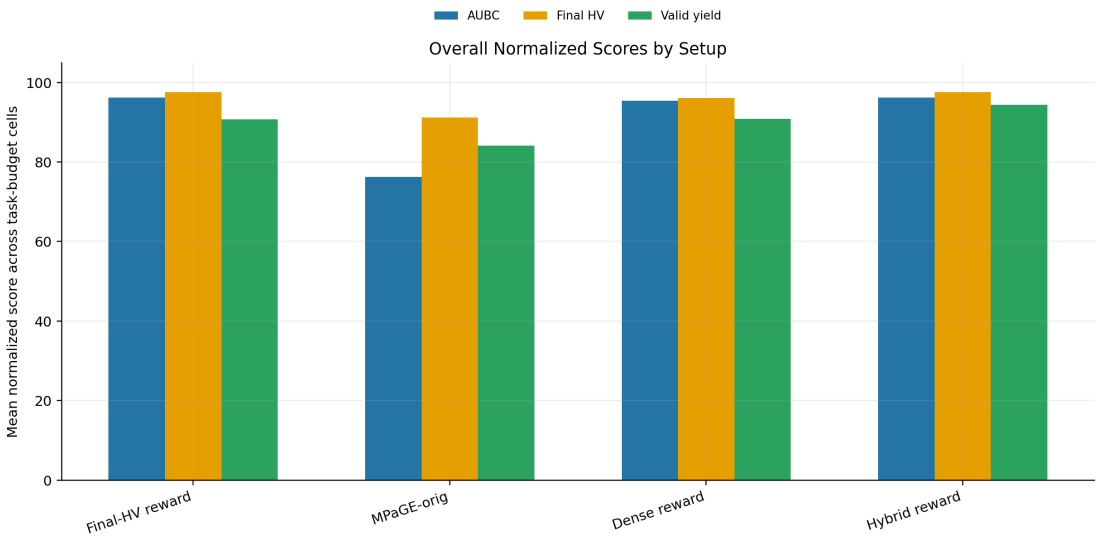


Figure 5.1: Overall normalized scores for AUBC, final HV, and valid heuristic yield across the four setups.

The strongest overall performance is achieved by Final-HV reward and Hybrid reward. Final-HV reward obtains a performance score of 96.9, the best mean final-HV rank, and the largest number of final-HV top-two cells. Hybrid reward has an almost identical performance score of 96.9, a marginally higher mean normalized AUBC score, and the best valid-yield score. This indicates that the hybrid reward is highly competitive, especially

when valid heuristic production is considered, but the final-HV-oriented reward remains slightly stronger in terminal population quality.

Dense reward is also competitive. Its performance score is 95.8, and it beats MPaGE-orig in every AUBC task–budget cell. This result is important because it shows that immediate HVI-style feedback remains useful when embedded in the finalized implementation. However, Dense reward has a lower final-HV score than Final-HV reward and Hybrid reward, suggesting that immediate HVI alone is not the best terminal-quality signal in the current result matrix.

MPaGE-orig has the weakest AUBC profile, with a mean normalized AUBC score of 76.2 and mean AUBC rank 3.94. Its final-HV score is closer to the reward-mode variants than its AUBC score, which suggests that the original MPaGE workflow can sometimes reach reasonable terminal populations, but it tends to do so later and less efficiently.

The overall comparison also reveals that the strongest method depends on which practical objective is emphasized. If the thesis objective is final managed-population quality, Final-HV reward is the most defensible default because it has the best final-HV mean rank. If the objective is valid candidate production, Hybrid reward is strongest. If the objective is only AUBC, Final-HV reward, Dense reward, and Hybrid reward are close enough that the result should be described as a family-level BMAB advantage rather than as a universal win by one reward mode. This nuance is important because it prevents overclaiming: the experiments support the proposed budget-aware framework strongly, but they do not show that one scalar reward is best in every possible sense.

5.5 Task-Level Patterns

Table 5.3 reports task-level mean normalized scores for AUBC and final HV. This table helps identify whether the overall conclusion is driven by one benchmark family or whether it is visible across tasks.

Table 5.3: Task-level mean normalized scores for AUBC and final outer hypervolume. Scores are normalized within each task–budget cell before averaging across budgets.

(a) AUBC normalized score

Setup	Bi-TSP	Tri-TSP	Bi-CVRP	Bi-KP
Final-HV reward	99.5	96.9	96.4	92.0
MPaGE-orig	88.7	77.5	66.1	72.5
Dense reward	98.3	98.4	95.2	89.7
Hybrid reward	98.1	99.3	94.8	92.7

(b) Final-HV normalized score

Setup	Bi-TSP	Tri-TSP	Bi-CVRP	Bi-KP
Final-HV reward	99.6	100.0	99.2	91.7
MPaGE-orig	99.6	99.8	87.3	78.1
Dense reward	99.1	99.7	94.1	91.5
Hybrid reward	98.9	99.8	93.1	98.5

The Bi-TSP results show that all methods are relatively close in final HV. MPaGE-orig even obtains a strong final-HV normalized score on Bi-TSP, indicating that the original workflow can find competitive terminal heuristic populations on this task. However, its AUBC score is lower than every BMAB-style method. This suggests that the main Bi-TSP advantage of BMAB is not necessarily a much better endpoint, but faster conversion of budget into a useful population.

Tri-TSP provides a clearer example of budget-trajectory improvement. The final-HV scores of all methods are high and close because the terminal populations reach similar outer-HV regions, but MPaGE-orig has a much lower AUBC score. This pattern means that the baseline often catches up late, while BMAB-style methods reach high outer HV earlier. The result is consistent with the intuition that tri-objective inner archives require broader early exploration and that adaptive cluster/operator allocation can help populate useful heuristic regions sooner.

Bi-CVRP is the task where the AUBC gap against MPaGE-orig is largest. The baseline AUBC normalized score is 66.1, while the BMAB-style methods are all near or above 94.8. This is practically meaningful because Bi-CVRP has stronger feasibility structure than TSP: route capacity constraints make many naive local modifications less useful. A scheduler that learns which semantic regions and operators produce valid, improving code can therefore have a larger effect.

Bi-KP shows a more mixed reward-mode pattern. Hybrid reward has the strongest final-HV normalized score on this task, while Final-HV reward remains competitive in

AUBC. This suggests that dense local feedback may be particularly helpful for the binary knapsack representation, where valid local changes can provide informative incremental improvement. At the same time, MPaGE-orig is substantially weaker in final HV on Bi-KP, which indicates that the finalized BMAB implementation improves more than only early progress on this benchmark.

Across tasks, the most stable conclusion is therefore not that the same setup wins every task, but that MPaGE-orig is consistently weak in AUBC while the three current BMAB reward modes remain competitive. This supports the thesis claim that adaptive LLM-call allocation is useful across multiple combinatorial representations.

5.6 Pairwise Cell-Level Comparison

Pairwise task–budget wins give a direct view of how often one setup has a higher mean metric value than another setup. Table 5.4 reports these counts for AUBC and final HV.

Table 5.4: Pairwise task–budget wins across 16 cells. Each entry gives the number of cells in which the row setup has a higher mean metric value than the column setup.

(a) AUBC wins

Row setup	Final-HV	MPaGE	Dense	Hybrid
Final-HV reward	–	15	9	10
MPaGE-orig	1	–	0	0
Dense reward	7	16	–	9
Hybrid reward	6	16	7	–

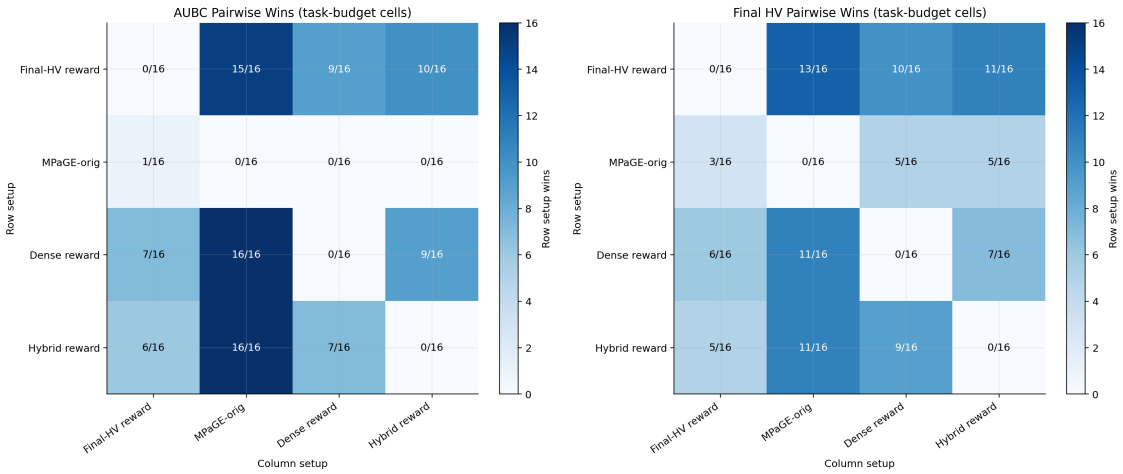
(b) Final-HV wins

Row setup	Final-HV	MPaGE	Dense	Hybrid
Final-HV reward	–	13	10	11
MPaGE-orig	3	–	5	5
Dense reward	6	11	–	7
Hybrid reward	5	11	9	–

For AUBC, every current BMAB-style setup is clearly stronger than MPaGE-orig. Final-HV reward wins 15 of 16 AUBC cells against MPaGE-orig. Dense reward and Hybrid reward each win all 16 AUBC cells against MPaGE-orig. This is the strongest evidence that budget-aware scheduling and the finalized BMAB implementation improve budget-integrated behavior relative to the original MPaGE workflow.

Among the three current reward modes, the AUBC comparison is close. Final-HV

reward wins 9 of 16 cells against Dense reward and 10 of 16 against Hybrid reward. Dense reward wins 9 of 16 against Hybrid reward. These margins indicate that no reward mode uniformly dominates all others in AUBC. The final-HV comparison is more favorable to Final-HV reward: it wins 10 of 16 cells against Dense reward and 11 of 16 against Hybrid reward. Therefore, Final-HV reward is the most defensible official default when terminal population quality is weighted together with budget efficiency.



(a) AUBC (b) Final HV

Figure 5.2: Pairwise task–budget win matrices for AUBC and final hypervolume.

5.7 AUBC Analysis

Figure 5.3 shows mean AUBC by task and budget. The central pattern is that all BMAB-style methods substantially outperform MPaGE-orig in budget-integrated performance. The gap is especially visible in Tri-TSP and Bi-CVRP, where the baseline tends to improve later in the budget trajectory. Since AUBC integrates the whole curve, delayed improvement is penalized even when final hypervolume eventually becomes competitive.

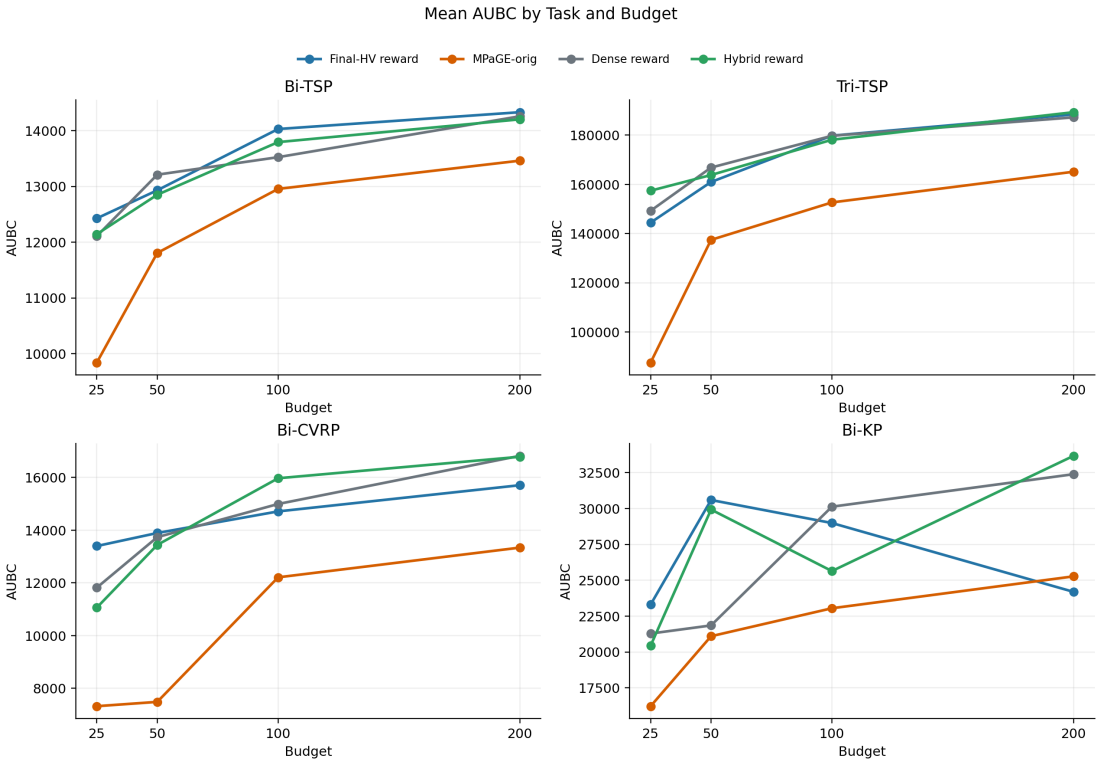


Figure 5.3: Mean AUBC by task and budget for the four final setups.

Final-HV reward obtains the best AUBC value in eight cells and is among the top two in eleven cells. Dense reward obtains four AUBC best cells and twelve top-two cells, while Hybrid reward obtains four AUBC best cells and nine top-two cells. These counts explain why their normalized AUBC scores are close. The difference between the reward modes is therefore not a simple win/loss story; rather, the reward modes trade off early dense feedback against final-population alignment.

MPaGE-orig obtains no AUBC top-two cells. This is an important result because it is robust to task and budget variation in the current matrix. The result does not mean that MPaGE-orig never generates useful heuristics. Instead, it means that under the project harness and finite call budgets, it reaches useful heuristic-population hypervolume later than the BMAB-style methods.

The AUBC result is particularly important at low and medium budgets. When $B = 25$ or $B = 50$, the number of generation opportunities is small, so early scheduling decisions have a large effect on the integral. Spending calls on unproductive semantic regions or on operators that currently produce invalid children can leave the curve flat for a substantial fraction of the run. BMAB-style scheduling mitigates this by using reward feedback after each generated child. The advantage is therefore structural: the controller can redirect calls during the run, whereas a fixed policy must wait for the next predetermined step.

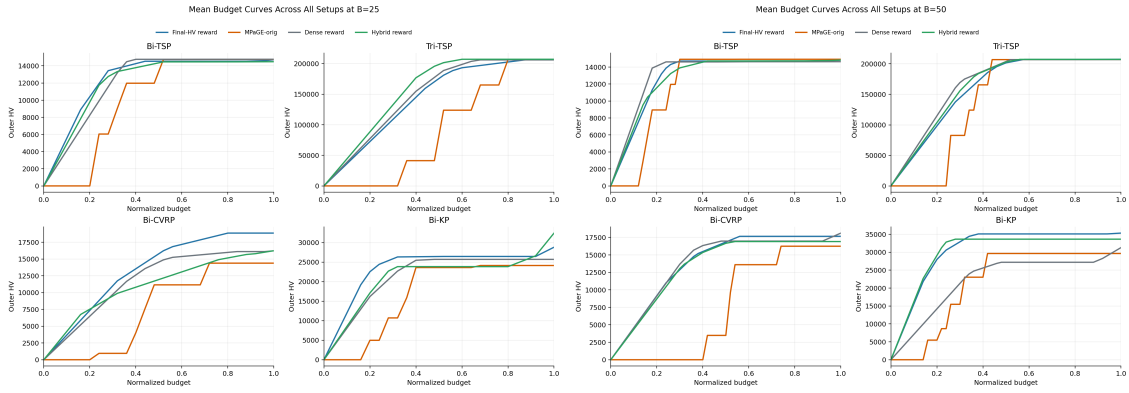
At larger budgets, the interpretation changes slightly. With $B = 100$ or $B = 200$, all

methods have more opportunities to recover from early poor choices. Consequently, final HV can become close even when AUBC remains different. This is why AUBC and final HV should be read together. A method that is worse in AUBC but similar in final HV is not necessarily incapable; it may simply require more budget to reach the same region. For LLM-driven heuristic design, that delay is itself an important cost.

The budget levels also change the effective learning problem faced by the bandit. At $B = 25$, a run may contain only a few meaningful decision rounds after initialization and clustering overhead. At $B = 200$, the controller has enough observations to revise its beliefs about operators and clusters several times. This creates a tension between early exploration and late exploitation, which motivates the budget-annealed cluster UCB term. Larger budgets also use larger managed populations, so validity diagnostics remain relevant: a larger population can help only if the method generates enough valid and useful programs to fill it.

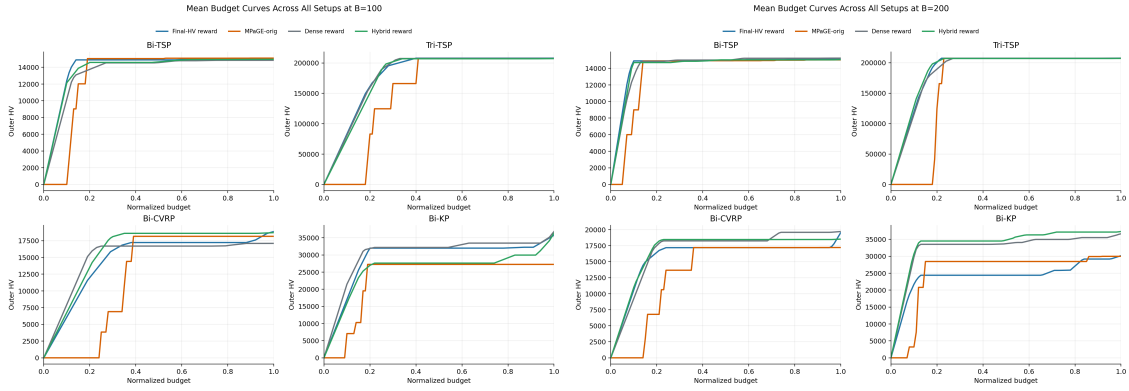
5.8 Budget-Curve Behavior

The budget curves in Figure 5.4 explain the AUBC results. The BMAB-style methods usually increase outer hypervolume early, often during or shortly after initialization. MPaGE-orig can exhibit a longer initial flat region, especially in Tri-TSP, Bi-CVRP, and Bi-KP. This delayed increase reduces AUBC even if the final value later approaches the BMAB-style methods.



(a) $B = 25$

(b) $B = 50$



(c) $B = 100$

(d) $B = 200$

Figure 5.4: Mean outer hypervolume curves over normalized budget for the four final setups.

Some curves become nearly flat after an early rise. This should not be interpreted as a plotting error. The profiler records curve points when the managed heuristic population changes at generation boundaries and at final pending-offspring flushing. Once the population reaches a high-HV region, later generated heuristics may not improve the managed Pareto population enough to change the outer hypervolume. The zoomed BMAB-style budget curve in Figure 5.5 makes these smaller differences easier to see without MPaGE-orig's early zero region dominating the scale.

The shape of a budget curve contains more information than the final point. A steep initial rise means that initialization and early generations found useful heuristics quickly. A long flat segment means that calls were spent without changing the managed population's outer HV. A late jump means that the method eventually discovered a useful heuristic but only after a period of low return. Since AUBC integrates all of these behaviors, it rewards early and sustained improvement rather than only endpoint quality.

This interpretation also explains why the y-value at normalized budget 1 in a budget-curve plot should be read carefully. The curve endpoint corresponds to the mean final outer HV only when the curve is aggregated and plotted using the same per-run final points and the same task–budget grouping as the final-HV summary. Differences can appear when

curves are averaged over interpolated budget positions, anchored at $(0, 0)$, or plotted with task-level grouping that differs from the final-HV bar chart. In this thesis, the generated overview figures were regenerated after the rerun correction so that the reported AUBC, final HV, and budget curves use the same current result artifacts.

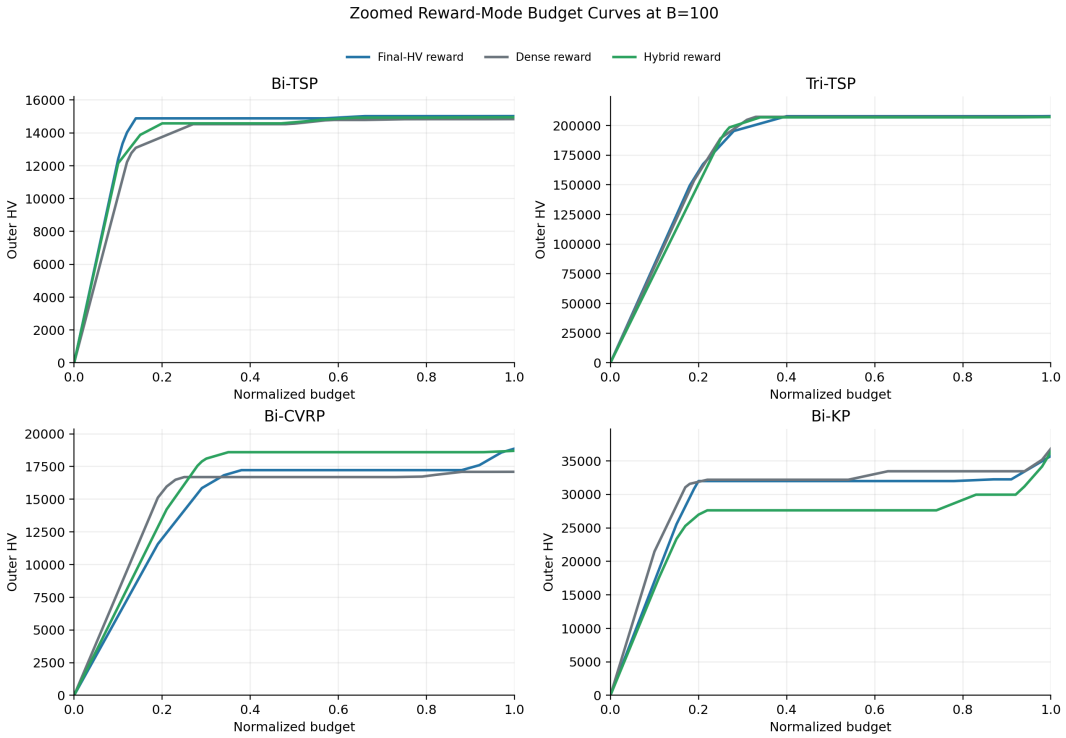


Figure 5.5: Zoomed BMAB-style budget curves at $B = 100$, excluding MPaGE-orig to make differences among Final-HV reward, Dense reward, and Hybrid reward more visible.

5.9 Final Hypervolume Analysis

Final hypervolume provides a different view from AUBC. Figure 5.6 shows that Final-HV reward has the strongest terminal-quality profile overall: it obtains the best final-HV rank, six best cells, and twelve top-two cells. It also wins 13 of 16 final-HV cells against MPaGE-orig. This supports the final-HV-oriented reward design as the safest default when the final managed population is the main concern.

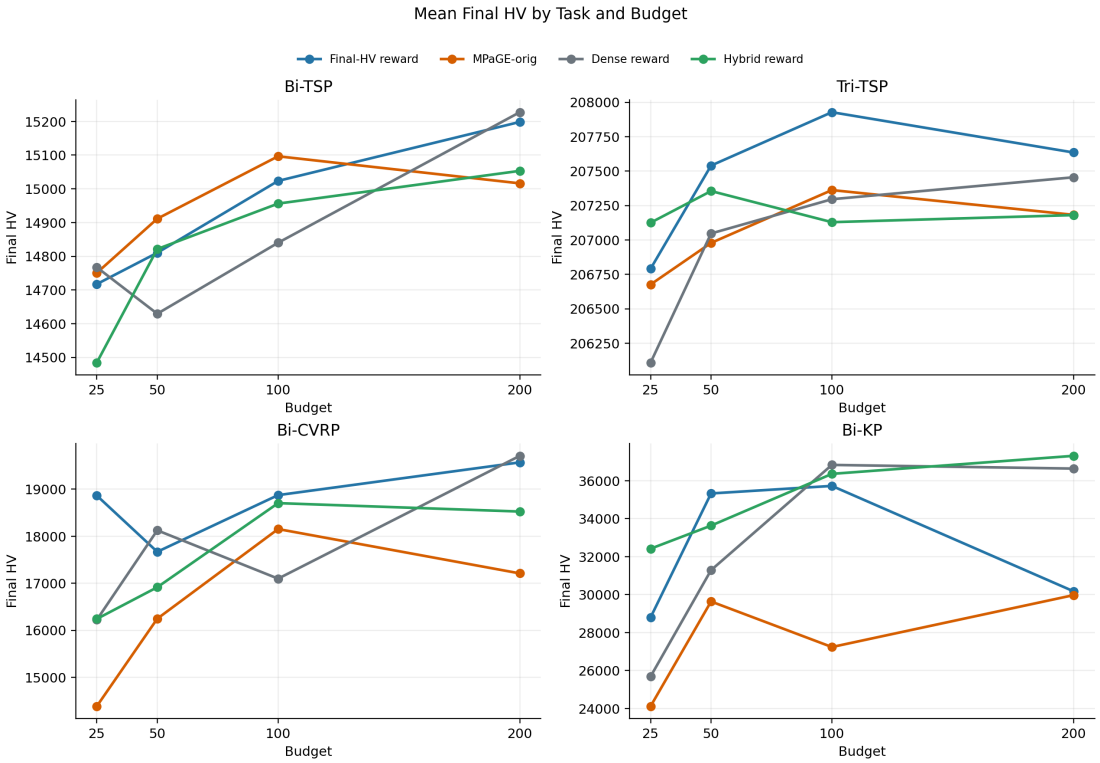


Figure 5.6: Mean final outer hypervolume by task and budget for the four final setups.

The final-HV results are nevertheless more mixed than AUBC. MPaGE-orig is weak in AUBC but still has two best final-HV cells. Dense and hybrid reward are also competitive in several final-HV cells. Therefore, the main advantage of the BMAB-style methods is not simply that they always end with much larger terminal HV. The more precise conclusion is that they improve the search trajectory and usually maintain or improve terminal population quality. This distinction is important because final-HV-only analysis would underestimate the value of budget-aware scheduling.

The trade-off plot in Figure 5.7 summarizes this relationship. Final-HV reward and Hybrid reward occupy the strongest region, with high normalized AUBC and high normalized final HV. Dense reward is slightly lower in final HV, while MPaGE-orig is separated mainly by its lower AUBC score.

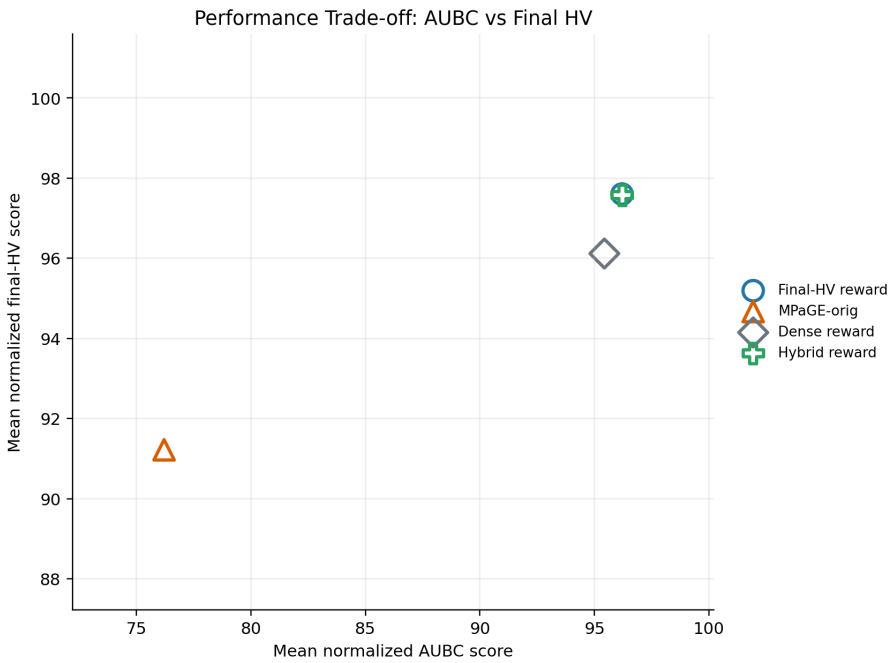


Figure 5.7: Trade-off between mean normalized AUBC score and mean normalized final-HV score.

The final-HV results support two methodological points. First, aligning the reward with final managed-population HV is useful. Final-HV reward has the best final-HV mean rank and more final-HV top-two cells than the reward variants. This is consistent with the design rationale: if the bandit is trained on the same population-level effect that is reported at the end, its updates are more directly aligned with terminal quality.

Second, final HV is not sensitive to all forms of wasted budget. If a method spends many calls early without improvement but eventually reaches a strong front, final HV can still look good. This explains why MPaGE-orig is less separated in final HV than in AUBC. For thesis evaluation, this distinction matters because a committee reader might otherwise ask why the proposed method is necessary when the final endpoints are sometimes close. The answer is that the proposed method changes the budget trajectory, not only the terminal snapshot.

The trade-off between AUBC and final HV also suggests different deployment preferences. If LLM calls are expensive or runs may be interrupted, AUBC should be prioritized. If only the final artifact matters and the full budget is guaranteed, final HV receives more weight. Final-HV reward is a strong default because it performs well under both views, whereas Hybrid reward is attractive when valid-yield and early progress are emphasized.

5.10 Reward-Mode Interpretation

The Final-HV reward, Dense reward, and Hybrid reward setups isolate the role of the main quality signal while keeping the finalized implementation otherwise unchanged. Dense reward provides immediate-HVI feedback: it rewards local hypervolume improvement against the current heuristic front. Final-HV reward evaluates the candidate by its effect after managed-population selection, aligning the bandit feedback more directly with the final reported population. Hybrid reward averages these two signals.

The results show that immediate dense feedback remains useful. Dense reward reaches a mean normalized AUBC score of 95.4 and beats MPaGE-orig in all AUBC cells. This suggests that short-horizon HVI feedback is informative for early progress. However, its final-HV score of 96.1 is lower than both Final-HV reward and Hybrid reward. This supports the methodological concern that an immediate HVI reward can encourage locally useful candidates without fully aligning with the managed final population.

Hybrid reward has the highest valid-yield score and the highest mean normalized AUBC score by a very small margin. This implies that combining dense feedback with final-population feedback can improve the rate at which valid and useful heuristics are produced. Nevertheless, Hybrid reward does not surpass Final-HV reward in final-HV rank or pairwise final-HV wins. Final-HV reward is therefore better justified as the default when both budget trajectory and final population quality must be considered.

The reward-mode comparison is also useful because it separates two kinds of improvement. The first kind is implementation-level improvement: pending-offspring flushing, final-call cluster guarding, front-aware priors, and weighted parent selection make the search process more faithful to the final-HV objective. The second kind is reward-signal improvement: the scalar feedback given to the bandit changes which arms are reinforced. Since `dense_reward` and `hybrid_reward` keep the finalized implementation, their results show that the implementation fixes are not dependent on a single reward variant.

The dense reward has an intuitive advantage: it gives immediate feedback whenever a candidate expands the current front. This can be helpful early because the front is sparse and many improvements are possible. Its weakness is that immediate HVI may reward candidates that improve the current front but are later discarded by the managed population cap or provide little benefit after survivor selection. The final-HV reward is more conservative, but it better matches the final metric. The hybrid reward sits between these two behaviors, and the results show exactly that: strong AUBC and validity, but not the strongest final-HV rank.

5.11 Validity Diagnostics

Figure 5.8 shows valid heuristic yield per 100 calls. Hybrid reward is strongest on this diagnostic: it has the best valid-yield rank, seven best cells, and fourteen top-two cells. This may help explain its strong AUBC performance. A setup that produces valid candidates more consistently has more opportunities to improve the heuristic population.

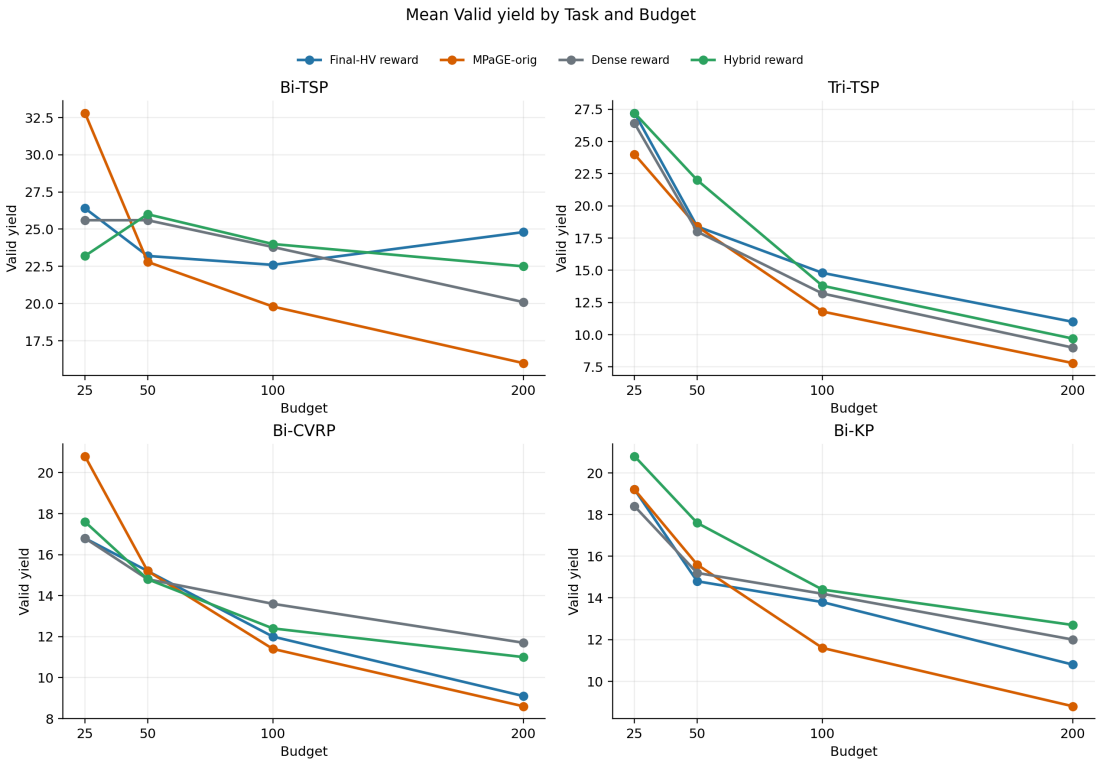


Figure 5.8: Mean valid heuristic yield per 100 calls by task and budget.

However, valid yield does not fully explain performance. Final-HV reward has a lower valid-yield score than Hybrid reward, yet it has better final-HV rank. This indicates that candidate quality and candidate placement in the managed Pareto population matter at least as much as raw valid count. In other words, producing more valid code is helpful, but not sufficient; the generated heuristics must also contribute useful trade-offs in the outer score space.

Figure 5.9 reports the invalid/null proxy rate. This figure is useful for diagnosing BMAB-style runs, but it should be interpreted cautiously when comparing against MPaGE-orig because the storage conventions differ. MPaGE-orig's diagnostic score appears very strong in the normalized diagnostic tables because its invalid/null information is recorded differently. For this reason, the thesis uses invalid/null rate as supporting evidence rather than as a primary ranking metric.

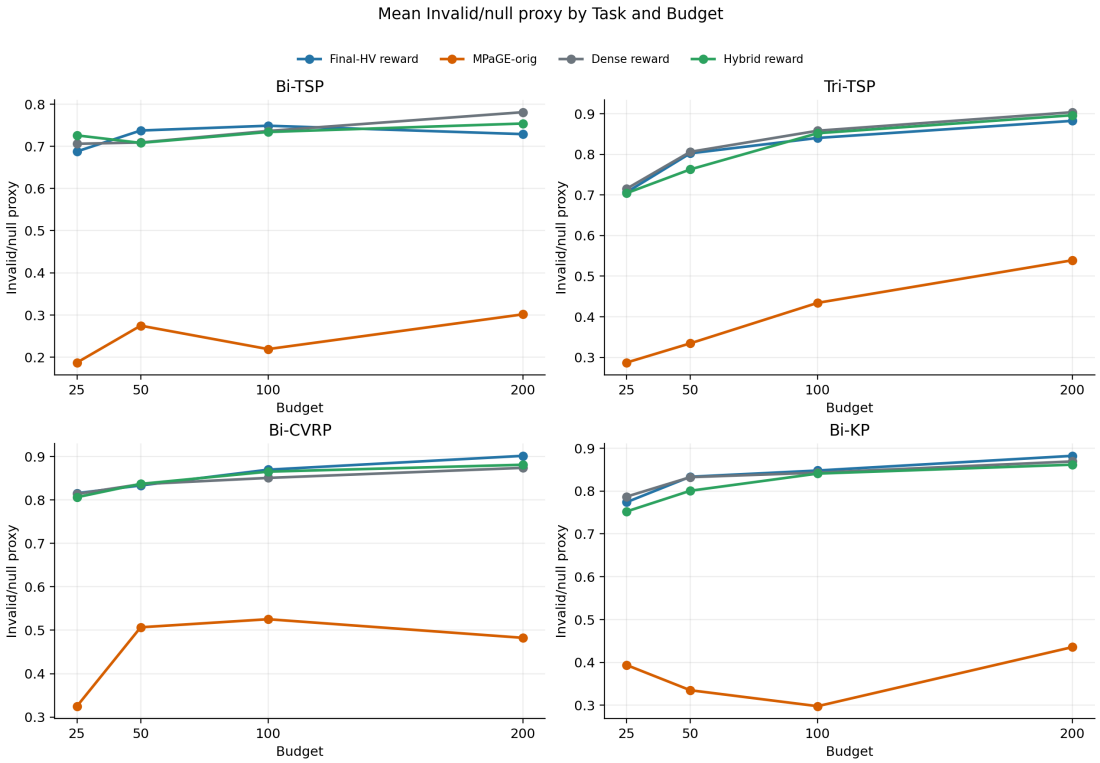


Figure 5.9: Mean invalid/null proxy rate by task and budget. Lower values are better, but storage conventions differ between BMAB-style runs and MPaGE-orig.

The validity diagnostics help avoid a simplistic interpretation of the results. If the best-performing method were simply the one that generated the most valid code, then valid-yield ranking would match AUBC and final-HV ranking. It does not. Hybrid reward is strongest in valid yield, but Final-HV reward is strongest in final-HV rank. This means that the search controller must balance at least two factors: producing executable candidates and producing candidates that occupy useful positions in the outer Pareto score space.

The invalid/null diagnostic also reveals a measurement limitation. Because the result formats differ between BMAB-style runs and MPaGE-orig, the normalized invalid/null score cannot be used as a fair standalone ranking across all four setups. This is why the thesis focuses on AUBC and final HV for performance claims. Validity metrics are still useful internally: they can identify cells where an unexpectedly weak AUBC value may be caused by too few valid offspring rather than by poor selection among valid offspring.

5.12 Rerun Cell and Result Consistency

During result inspection, the Dense reward Bi-CVRP $B = 25$, seed-2029 cell was rerun because the previous artifact had positive final HV but AUBC equal to zero. The updated artifact used in the consolidated overview produces AUBC 12725.91, final HV 16315.27,

and four valid heuristics. Table 5.5 shows the resulting per-seed values for that cell.

Table 5.5: Per-seed dense-reward results for the Bi-CVRP $B = 25$ cell after rerunning seed 2029.

Seed	AUBC	Final HV	Valid	Gen. attempts	Invalid proxy
2025	12909.17	16302.48	5	22	0.773
2026	12861.59	17863.32	4	23	0.826
2027	13136.16	17751.57	4	22	0.818
2028	7474.50	12887.06	4	24	0.833
2029	12725.91	16315.27	4	22	0.818

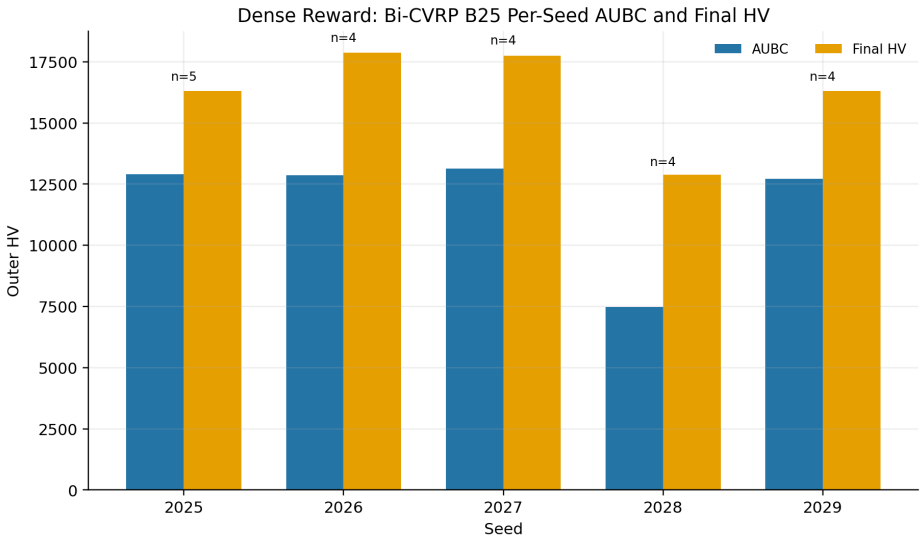


Figure 5.10: Dense-reward Bi-CVRP $B = 25$ per-seed AUBC and final HV after rerunning seed 2029.

The rerun removes the zero-AUBC anomaly and makes seed 2029 consistent with the other dense-reward Bi-CVRP $B = 25$ seeds. The consolidated overview and thesis tables were regenerated after this rerun, so the reported dense-reward means in this chapter use the corrected artifact.

This rerun is a useful example of why budget-curve artifacts are necessary. A final-HV value alone would not reveal whether the trajectory was recorded correctly. A positive final HV with zero AUBC is suspicious because AUBC should integrate the same curve whose final point supplies final HV. By inspecting and regenerating the affected cell, the analysis ensures that the aggregated dense-reward results do not contain a known inconsistency.

5.13 Limitations of the Evidence

The result matrix is complete for the four setups analyzed here, but each task–budget cell still contains only five seeds. This is sufficient for observing broad empirical trends, but it is not sufficient for strong statistical claims. The consolidated overview provides normalized scores, ranks, and pairwise win matrices; it does not provide a complete all-pairs inferential test suite for every method pair and metric. Therefore, this chapter emphasizes practical effect patterns rather than formal statistical significance.

Dense reward and Hybrid reward are reward-mode ablations of the current implementation, not separate systems with different schedulers. Therefore, differences among the BMAB-style setups should be interpreted as differences in bandit feedback, not as evidence for completely separate architectures.

The repository still does not provide token usage, API latency, monetary cost, GPU usage, or memory consumption for the reported runs. No information was found in the project source code or documentation for these measurements. Consequently, the budget model in this thesis remains call-based.

Finally, all reported HV and AUBC values are outer heuristic-population metrics. They are meaningful for comparing heuristic-design processes under the `mpage_bmab` harness, but they are not direct reproductions of the normalized inner-solver metrics reported in the original MPaGE paper.

Another limitation is that the experiment matrix compares reward modes but does not yet fully isolate every architectural component. The codebase contains ablation hooks for Page–Hinkley, diversity reward, cluster-only control, operator-only control, and budget annealing, but complete result matrices for all of these variants are not part of the current thesis evidence. Therefore, the chapter can conclude that the finalized BMAB-style methods work well as a family, but it cannot assign a precise causal contribution to every individual component.

The analysis also remains quantitative rather than qualitative. It does not inspect the generated heuristic code to identify recurring algorithmic motifs. Such an analysis could reveal whether BMAB scheduling discovers genuinely different heuristic families or mainly reallocates budget among similar local variants. The saved samples make this possible, but it is outside the present scope.

5.14 Experimental Conclusion

The current result artifacts support three main findings. First, the BMAB-style methods are substantially stronger than MPaGE-orig in AUBC. This indicates that budget-aware schedul-

ing and the finalized implementation improve how quickly useful heuristic populations are obtained. Second, Final-HV reward is the strongest default when AUBC and final HV are considered together: it is effectively tied with Hybrid reward in overall performance score and has the best final-HV rank. Third, the reward-mode comparison reveals a meaningful trade-off. Dense and hybrid rewards provide strong early feedback and valid-yield behavior, while the final-HV reward gives the most reliable terminal population quality.

Overall, the experiments support the thesis claim that LLM-call allocation is an algorithmically important part of MPaGE-style heuristic design. The gains are not explained solely by producing more valid programs. Rather, the evidence suggests that adaptive selection of operators, semantic regions, and reward signals changes the budget trajectory of heuristic-population search.

Chapter 6

Conclusion and Future Work

6.1 Summary

This thesis analyzed `mpage_bmab`, a budget-aware extension of MPaGE for LLM-based heuristic design in multi-objective combinatorial optimization. The project preserves MPaGE's core generative and evaluative mechanisms, including SEMO-style archive updates, Pareto Front Grid selection, semantic clustering, prompt-based heuristic generation, and benchmark evaluation. It extends this foundation with an adaptive scheduling layer that treats LLM calls as a scarce resource.

The proposed BMAB-LLM method can be summarized as follows. Each generator or clusterer invocation is charged by a hard `BudgetTracker`. An operator-level UCB controller chooses between mutation and crossover. A cluster-level budgeted UCB controller selects semantic cluster–operator arms within each generation. Page–Hinkley drift detection resets stale cluster statistics when reward behavior changes. A bounded reward function converts heuristic-evaluation outcomes into bandit feedback using final managed-population hypervolume gain, diversity gain, rank smoothing, and invalid-code penalties. The profiler records budget curves, AUBC, bandit logs, and final population artifacts for analysis.

At a broader level, the project shows that MPaGE-style heuristic design can be strengthened by making LLM-call allocation adaptive. MPaGE already provides a productive foundation through PFG selection, semantic clustering, and prompt-based variation. The contribution of `mpage_bmab` is to place these mechanisms under a controller that learns which operators and semantic regions deserve additional budget during the run.

6.2 Main Contributions

The first contribution is the formulation of LLM-based heuristic design as a budget-constrained sequential decision problem. Rather than treating the budget merely as a

stopping condition, the project makes budget consumption part of the algorithmic state and evaluates progress over the entire budget trajectory through AUBC.

The second contribution is a two-level bandit scheduler for MPaGE-style heuristic generation. The operator bandit learns long-term preferences between mutation and crossover, while the cluster bandit adapts to the short-lived semantic clusters constructed in each generation. This structure is well matched to the project setting: operators are stable across the run, whereas cluster identities are local and non-stationary.

The third contribution is the integration of non-stationarity handling through Page–Hinkley drift detection. Because a semantic cluster can become less useful after repeated exploitation, resetting stale cluster statistics is a practical mechanism for avoiding persistent overconfidence in outdated reward estimates.

The fourth contribution is the final-HV-oriented reward and population-handling design. The finalized method rewards the actual generated child, aligns the quality signal with the managed final-population hypervolume, prevents the last affordable call from being spent only on clustering, incorporates front-aware cluster priors, weights parents by PFG membership, and flushes pending valid offspring into the final population before result writing.

The fifth contribution is an experimental and analysis pipeline that records not only terminal results but also the process by which budget is consumed. Artifacts such as `budget_curve.json`, `aubc.json`, `bandit_log.json`, validity summaries, and the generated thesis figures under `mpage_bmab/thesis/figures/` support analysis of both final quality and budget efficiency.

6.3 Main Findings

The current result set provides strong practical evidence that BMAB-style scheduling improves budget efficiency over the original MPaGE workflow. Across the four-setup matrix, the Final-HV reward method wins 15 of 16 AUBC task–budget cells against MPaGE-orig, while Dense reward and Hybrid reward each win all 16 AUBC cells against MPaGE-orig. The largest qualitative differences appear in tasks where the baseline budget curve rises later, particularly Tri-TSP, Bi-CVRP, and Bi-KP.

The evidence for final hypervolume is also favorable to the Final-HV reward method, but it remains more nuanced than AUBC. This setup has the best mean final-HV rank, six best final-HV cells, and twelve top-two final-HV cells. It also wins 13 of 16 final-HV cells against MPaGE-orig. Nevertheless, Dense reward, Hybrid reward, and occasionally MPaGE-orig remain competitive in some terminal cells. Therefore, the correct conclusion is not uniform terminal domination, but improved budget trajectory with generally strong terminal quality.

The reward-mode comparison clarifies an important trade-off. Hybrid reward obtains

the best valid-yield profile and a slightly higher normalized AUBC score than Final-HV reward, whereas Final-HV reward has the strongest final-HV rank. Dense reward remains competitive but is weaker than Final-HV reward in final-HV score. These results support the final-HV-oriented reward as the safest official default while showing that dense and hybrid reward signals remain useful for early progress and valid candidate production.

The task-level analysis gives a more nuanced view. On Bi-TSP, several methods reach similar final-HV values, so the BMAB advantage is expressed mainly through better budget trajectory. On Tri-TSP, final-HV values are also close, but BMAB-style methods reach high outer HV earlier. On Bi-CVRP, the AUBC gap against MPaGE-orig is especially large, suggesting that adaptive scheduling is valuable when route feasibility and structured local moves make generation more fragile. On Bi-KP, hybrid reward is particularly competitive, which suggests that dense local feedback may be useful for binary item-selection representations.

The most accurate empirical conclusion is therefore careful rather than absolute. The proposed method should not be described as uniformly dominating every setup on every metric. It should be described as consistently improving budget-integrated performance while maintaining strong terminal quality. This interpretation matches the research objective more closely than a final-HV-only summary would.

6.4 Limitations

The thesis is limited by the available experimental evidence. Each task–budget cell contains five seeds, which is too small for strong statistical conclusions under a two-sided Wilcoxon signed-rank test. The reported trends are broad and consistent, but they should be interpreted as empirical evidence rather than definitive statistical proof.

The current complete result matrix includes the MPaGE baseline, the official Final-HV reward method, and two reward-mode variants. However, it does not include complete matrices for all component ablations. The codebase implements additional variants, including `no_budget_anneal`, `no_ph`, `no_diversity`, `op_only`, and `cluster_only`. Without complete result matrices for these variants, the thesis cannot isolate the causal contribution of every scheduler component.

The repository does not provide hardware specifications, GPU usage, memory usage, API latency, token-level cost, or monetary cost for the reported runs. No information was found in the project source code or documentation for these measurements. Consequently, the current budget model counts LLM calls rather than modeling heterogeneous token costs or real-time latency.

Finally, the reported HV and AUBC values are outer heuristic-population metrics.

They are meaningful for comparing budget-aware heuristic-design processes under the project harness, but they are not a direct reproduction of all inner-task solution-quality tables from the original MPaGE paper.

The thesis is also limited by the absence of qualitative code analysis. The saved generated samples make it possible to inspect recurring heuristic motifs, invalid-generation patterns, or semantic differences between clusters, but the current study focuses on quantitative outer-population metrics. Such qualitative analysis would be useful for explaining why particular reward modes or task families behave differently.

6.5 Future Work

The most immediate future direction is to run a complete component ablation matrix. The reward-mode comparison is now available for `full`, `dense_reward`, and `hybrid_reward`, but the effects of Page–Hinkley drift detection, diversity reward, operator-only scheduling, cluster-only scheduling, and budget annealing still require complete controlled sweeps.

A second direction is to increase the number of seeds. More seeds would improve the power of statistical tests and make it easier to distinguish robust effects from noise caused by stochastic LLM generation, random initial archives, and generated heuristic randomness.

A third direction is to extend the budget model from uniform call cost to token- or latency-aware cost. In practical LLM deployments, prompts and completions can differ substantially in length, and different models can have different price and latency profiles. A token-aware budgeted bandit would better reflect deployment constraints.

A fourth direction is to log and analyze invalid-generation failures more precisely. Separating syntax errors, missing function definitions, runtime exceptions, infeasible outputs, and timeouts would support targeted prompt engineering and repair strategies. Such diagnostics could also clarify whether validity improvements or reward allocation contribute more to AUBC gains.

Finally, the search controller could be extended with higher-level planning mechanisms, such as prompt repair, memory over successful heuristic motifs, or structured exploration over operator sequences. These extensions should be evaluated carefully because any additional LLM calls must justify their cost under the same AUBC-oriented budget criterion.

A sixth direction is adaptive reward scheduling. The current results show that dense reward is useful for early progress and that final-HV reward is strongest for terminal quality. This suggests a natural schedule in which the controller uses more dense or hybrid feedback early and gradually shifts toward final-HV feedback as the remaining budget decreases.

A seventh direction is qualitative heuristic analysis. Since the repository stores gen-

erated samples, future work can cluster or manually inspect the generated code to identify recurring local-search structures. This could answer questions that pure HV metrics cannot answer, such as whether successful heuristics share common move operators, whether invalid code arises from specific prompt patterns, or whether crossover actually combines useful motifs from two parents.

An eighth direction is token-aware and cost-aware budgeting. The current budget model counts LLM calls uniformly, but real model usage depends on prompt length, completion length, latency, and model pricing. A future cost-aware controller could treat mutation, crossover, and clustering as actions with variable monetary or token costs. This would make the budgeted-bandit formulation closer to real deployment conditions.

6.6 Final Remarks

`mpage_bmab` demonstrates that budget allocation is a substantive algorithmic issue in LLM-driven heuristic design. By adding a two-level bandit scheduler, hard budget accounting, drift detection, and budget-curve profiling to MPaGE, the project turns a fixed-budget search loop into an adaptive resource-allocation process. The current experiments indicate that this design substantially improves budget-integrated performance, especially on more complex benchmark families. Although further ablations and larger seed counts are required for stronger causal and statistical claims, the project provides a coherent foundation for studying Generative AI heuristic design under realistic cost constraints.

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